MILITARY SPECIFICATION

MANUALS, TECHNICAL: SCHEMATIC BLOCK DIAGRAMS (SBD)

MAINTENANCE DEPENDENCY CHARTS (MDC)

(For Aircraft, Missile/Space, Ground Communication-Electronic-Meteorological (C-E-M) and Related Equipment, Site, Systems and Facilities)

1. SCOPE

1.1 This specification covers the detail requirements for preparation of trouble-shooting data in the form of schematic block diagrams (SBD) and maintenance dependency charts (MDC), in lieu of the usual text, illustration, and tabular format for applicable aircraft, missile/space, ground communication-electronic-meteorological (C-E-M) and related equipment, sites, systems and facilities. Data shall be provided by the SBD/MDC technique only when contractually specified.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-M-38784	Manuals, Technical: General Requirements for Preparation of
MIL-P-38790	Printing Production of Technical Manuals: General Requirements of
MIL-M-38800	Manuals, Technical: Organizational Maintenance (For Aircraft)

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preparation. Schematic block diagrams and maintenance dependency chart manuals are considered nonstandard publications. The requirements of MIL-M-38784 are applicable in connection with the style and format of the cover page, list of effective pages, table of contents, introduction, text, alphabetical index, marginal copy, security classification markings, warnings, cautions and notes, figure titles, figure numbering, changes, revisions, artwork and size of type. The applicable documents (except MIL-M-63000) of MIL-M-38784 are also applicable hereto Further, the negative, reproduction assembly sheet, and printing requirements of MIL-P-38790 are applicable.

3.1.1 Presentation. Reproducible camera-ready copy shall be prepared in 10 3/4 by $16\frac{1}{2}$ inch size, or for reduction to that size. The maximum printing area (image size) for each page shall be $9\frac{1}{4}$ by $15\frac{1}{4}$ inches. The schematic block diagrams shall be prepared for reproduction on the lower page. The keyed text will be placed on the facing page rather than on the schematic block diagram. Pages shall be laid out for head to foot printing allowing for binding margins at the top of odd numbered pages and bottom of even numbered pages. Publication numbers shall be in the upper right corner and page numbers shall be in the lower right hand corner

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of all pages (head and foot alike).

3.1.2 Color. Color shading may be used in selected applications if it does not significantly increase the printing costs.

3.1.3 Style and coverage. Sentences need not be complete when the noun is clearly understood. Sentences should begin with a verb and should describe "what" the circuit does: e.g. "(generates 4V, 10usec pulse with a PRF on 4kh)". Articles (a, an, the) may be dropped to conserve space. If timing of a circuit needs to be described, the sentence shall begin with a "when" clause followed by a main clause beginning with a verb: e.g. "When energized, causes short pulse operation." If the purpose of the circuit is not obvious, a "why" statement shall also be included.

3.2 Arrangement. Manuals shall be arranged as follows:

Cover Page

List of Effective Pages

Table of Contents

Introduction

Input Conditions Page

Locator Page

Maintenance Dependency Charts

Keyed Text

Schematic Block Diagrams

Supplemental Data

Alphabetical Index

Back Cover

3.2.1 Cover page. The requirements of MIL-M-38784 are applicable except that the information shall be centered and within a 9¼ by 15¼ inch area. (See figure 1.)

3.2.2 List of effective pages. The requirements of MIL-M-38784 are applicable including the size of type. (See figure 2.)

3.2.3 Table of contents. The requirements of MIL-M-38784 are applicable. The table of contents shall be the only information placed on this page and the titles and page numbers of each schematic block diagram and each maintenance dependency chart shall be spaced proportionally within the entire image area (9¼ by 15¼ inches). (See figure 3.)

3.2.4 Introduction. The requirements of MIL-M-38784 are applicable. For technical orders covering a complex system/equipment, the basic introduction shall cover the system/equipment in general, indicating the overall relationship of major functions. Each major function shall have a brief introduction explaining content usage as may be required. The entire image area (9¼ by 15¼ inches) shall be used to explain the contents and usage of the manuals. Text columns shall be 3½ by 9¼ inches with 3/4 inches between columns. It is optional whether the Introduction backs up the Table of Contents or is placed on a separate sheet. (See figure 4.)

3.2.5 Input conditions page. The input conditions page shall include all the information required to prepare the system/equipment for the maintenance function. For manuals covering a complex system/equipment, a separate input condition page shall be prepared for each major function, i.e. Antenna Control/Drive Function,

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System Interlock Function, etc. When more than one sheet (page) is required, additional sheets shall be identified as (sheet 1 of 4), (sheet 2 of 4), etc, following the Input Condition page title. (See Figures 5 and 11). The following items, in the order indicated, shall be included in the input conditions page:

a. System Title

b. Applicability (Serial Numbers)

c. Supplies

d. Personnel Required

e. Special Tools and Test Equipment

f. Equipment Condition

g. Location Illustration for Input Conditions Page

3.2.5.1 System title. System titles shall specify the title of the system or subsystem for which the Schematic Block Diagrams and Maintenance Dependency Charts have been prepared.

3.2.5.2 Applicability. Applicability of the instructions to various equipment configurations shall be stated in accordance with MIL-M-38784. If the instructions apply to all configurations, the word "all" shall be entered.

3.2.5.3 Supplies. Expendable items and support materials shall be listed. Sealants, lubricants, replacement lamps, "maintenance in progress" labels, etc, are considered expendable items. The quantity of each expendable item required shall be given.

3.2.5.4 Personnel required. The minimum number of personnel required to effectively perform the maintenance shall be specified. Action requirements shall be summarized for each technician including location(s) and duties at the beginning of the function. Specialists (personnel competent in certain procedures, such as hydraulics, pneumatics, etc) required, will be listed.

3.2.5.5 Special tools and test equipment. For each procedure, identify Government approved AGE (test equipment, special tools, and ground handling equipment) applicable to system maintenance. Common type of tools such as screwdrivers, pliers, side cutters, soldering irons, etc, shall not be listed. Common types of test equipment such as voltmeters, signal generators, tube testers, etc, shall be listed. The special tools and test equipment shall be identified by standard nomenclature and the applicable MS, AN, AF, MIL, FSN or JAN part number.

3.2.5.6 Equipment conditions. Certain functions are dependent upon the system/equipment being in a given condition (e.g. aircraft must be on hydraulic jacks prior to removing shock strut). These required conditions must be stated. Any information pertinent to safety shall also be included and presented as a caution or warning. When the required conditions can be obtained by performing other organizational maintenance instructions, these instructions shall be indicated as prerequisites.

3.2.5.7 General information. Any notes, cautions, or warnings which pertain to the entire function shall be provided, such as: NOTE-local altitude must be known; WARNING - advise tower that test signals will be transmitted on a specified frequency. Notes, cautions, and warnings shall precede the step(s) to which they apply. If a Maintenance Support Information Manual is required, reference to the applicable publication number shall be made.

3.2.5.8 Location illustration for the Input Conditions Page. The location of all switches, circuit breakers, controls, and other items referred to or actuated by

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the procedural steps on the input conditions page shall be graphically illustrated. (See figures 5 and 11) This illustration shall be a part of the input conditions page and shall be in accordance with the following:

3.2.5.8.1 Presentation. The requirements of MIL-M-38784 are applicable. The entire image area shall be used to present the procedural information and companion illustration. The text on the input conditions page shall be columnar; $3\frac{1}{2}$ by $9\frac{1}{4}$ inches, with 3/4 inch between columns. Text may cover all the area except that occupied by the illustration(s) supporting the text. If insufficient space is available for the text, another input conditions page shall be started in the same format, with illustrations included as necessary to support only the text on that page.

3.2.6 Locator page. The locator page shall provide locator numerical index number and equipment/unit/segment name which will be referenced on the MDC and the schematic block diagram. The locator diagrams will locate systems, subsystems and components within the overall equipment at a level of detail sufficient to permit identification of the components. A locator page shall be prepared for each system or subsystem and shall be placed on the upper page (back of the preceding page) or start on an even numbered page facing the Maintenance Dependency Chart (MDC). (See figures 8 and 12.)

3.2.7 Maintenance Dependency Chart (MDC). A maintenance dependency chart is a chart which symbolically illustrates the dependency relationships of all procedural elements and all components within a system or subsystem. (See Figures 9, 13, 17 and 18) At least one MDC shall be provided for each complex troubleshooting task.

3.2.7.1 Organization. An MDC shall contain the following sections laid out as shown in Figures 9, 13, 17 and 18, with the content and format specified in the following paragraphs:

a. Headings

b. Procedure Column

c. Signal Values

d. Body

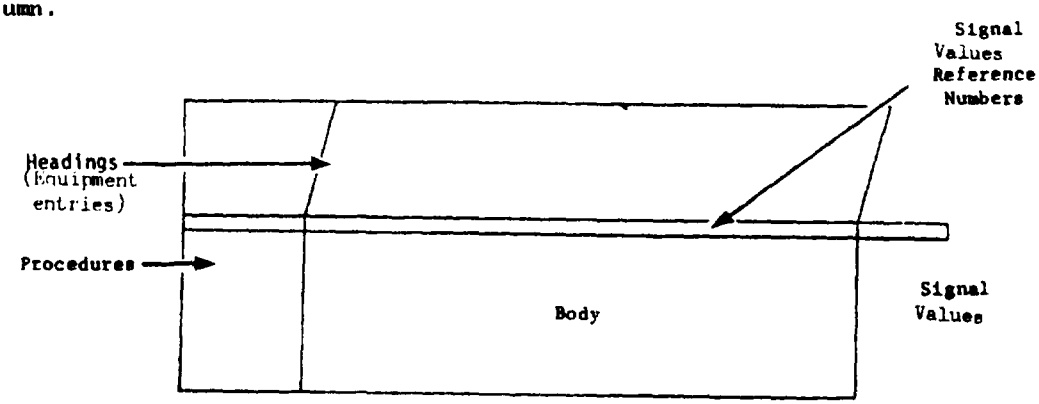
3.2.7.2 Headings. Heading information shall consist of the following:

a. Locator Information

b. Equipment Entries

c. Signal Value reference numbers

3.2.7.2.1 Locator information. The locator information shall be in the form of a callout number from the locator page which faces the MDC. This callout number identifies the location of the equipment component entered below it in the heading column.



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3.2.7.2.2 Equipment entries. The equipment entries shall include components and points of test. Components and points of test shall be identified with the same nomenclature used on the locator page.

3.2.7.2.3 Signal Value Reference number. The signal value reference number refers to a specified value for the signal which shall be contained in the signal values section, described in 3.2.8.4. A number shall appear only in a column where a signal is present for measurement. Numbers shall not be repeated since a signal at a specific point is to be considered unique. The numbers shall be Arabic and progress from left to right across the page.

3.2.7.3 Procedure column. The procedure column shall state all the procedural steps required to completely exercise all positions of all switches and operating controls through their full range, including all meaningful combinations of positions, in all equipment modes upon which they have an effect. Steps shall consist of control movements that produce observable results, and each result shall be specified in terms of the expected nominal reading and acceptable tolerance. Procedural steps shall cover situations where there are no external symptoms or self-test capabilities by using external test equipment for fault isolation or signal injection. External test equipment operating procedures shall not be included as an MDC procedural step.

3.2.7.3.1 Sequence of procedural steps. The sequence of steps shall be so identified that the first step energizes the system or subsystem and then determines the malfunctioning unit.

3.2.7.3.2 Symbols. Three basic symbols shall be used to show the interrelationships between the components shown on the schematic block diagram and their input and output signals. It shall be the contractor's responsibility to generate new symbols consistent with those specified herein. All new symbols shall be presented to the Procuring Agency for review and approval prior to publication of the manual. A key for the MDC symbols, used in each volume containing MDCs, shall be included in the Introduction.

a. Availability or output box. [A] An availability box shall be used to depict a resultant activity or availability produced by a single series dependency, and upon which other events and activities are dependent. The availability box shall be open and contain an "A" unless additional information about the availability exists. If additional information exists it shall be abbreviated or otherwise shortened as necessary to fit in an availability box. The following are acceptable exampes of additional information in availability boxes.

[LIT]	Indicator: LIT = Lighted	[STOP]	STOP = Stop Motor
[OUT]	OUT = Light off	[YES]	In conjunction with signal specification
[EN]	Energy application: EN = Energize	[NO]	This indicates availability condition
[DE]	DE = Deenergize	[IND]	Indicator (e.g., torquemeter)
[RUN]	Actuation (Motor): RUN = Run Motor		

b. ● Participation in a dependency chain by a functional segment.

Ⓒ Closed - The closed symbol applies to switches, valves, etc., and indicates that data flow is interrupted until the switch is operated.

⊖ Open - The open symbol applies to switches, valves, etc., and indicates that data flow is not interrupted until the switch is operated.

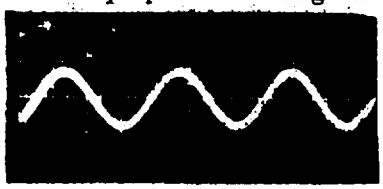
c. ▲ Dependency Marker - Used to indicate the dependency upon another event.

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3.2.7.4 Signal Values. This section of the MDC shall state the signal values that are referenced in accordance with 3.2.8.2.3. Signal values shall be stated in the following forms as applicable:

a. A white on black photograph or line drawing of waveforms showing the graticule and including the associated tolerances. Time per unit distance and voltage per unit distance must be specified in terms appropriate for the test instrument specified by the Special Tools and Test Equipment listing in the associated Input Conditions Page.

1ST VARIABLE 30 CPS AMPLIFIER V9
Connection Point: TP4
Vertical Sensitivity: 0.005 volts/cm X 10:1 probe
Signal Frequency: 30 cps Variable
Sweep Time: 10 ms/cm



b. Voltage, amperage, resistance, RPM, PSI, and other signal values with associated tolerances shall be stated in terms of function, range and display labeling of the test instrument specified by the Special Tools and Test Equipment listing in the associated Input Conditions Page. (See figure 18.)

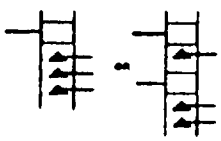
3.2.7.5 Body. The body of an MDC shall consist of a series of horizontal dependency lines containing specified symbols which relate a functional segment or group of functional segments to their associated inputs and outputs. The organization of these dependency lines in the chart body, called the dependency structure, shall portray the energy flow through the component depicted on the chart.

3.2.7.5.1 Dependency structures and diagramming techniques. A dependency line shall consist of dependency markers, participation dots and event boxes connected as follows:

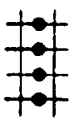


The symbols in dependency lines shall be ordered into one of two types of columns: the event/dependency column and the functional participation column.

The event/dependency column shall consist of event boxes and dependency markers depicted as follows:



The component column shall consist only of participation symbols, even though a component participation may be depicted more than once. It is indicated as follows:



3.2.7.5.2 Mapping methods and techniques. The following methods and techniques shall govern substantially all the basic mapping situations to be encountered. If situations arise which are not covered in the following paragraphs, the contractor shall augment the mapping schemes and record the augmented schemes for future use.

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a. Simple dependency line. The mapping of a component with one input and a single output results in a simple dependency line as follows:

Schematic Block Diagram



MDC



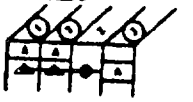
The mapping shows that the output event at 2 is dependent upon component z and the input event at 1.

b. Multiple input dependency line. A component with more than one input and a single output is mapped as a multiple input dependency line as follows:

Schematic Block Diagram



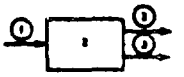
MDC



The mapping shows that the output event at 3 is dependent upon component z and the input events at 1 and 2.

c. Multiple output dependency line. A component having one input and multiple outputs shall be mapped as a multiple output dependency line, as follows:

Schematic Block Diagram



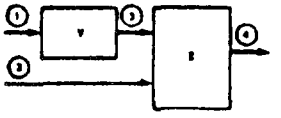
MDC



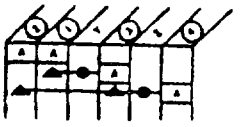
The mapping shows that both outputs at 2 and 3 are dependent upon component z and the input event at 1.

d. Dependency chain. A dependency chain consists of a series of interrelated dependency lines. The following example shows a dependency chain consisting of two dependency lines.

Schematic Block Diagram



MDC



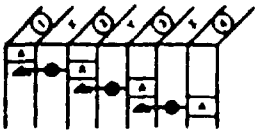
The last dependency line in this example shows that the output event at 4 is dependent upon component z and the events at 3 and 2.

e. Serial relationship. Serial relationships shall be shown as follows:

Schematic Block Diagram



MDC

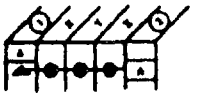


or

Schematic Block Diagram



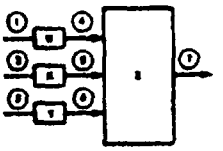
MDC



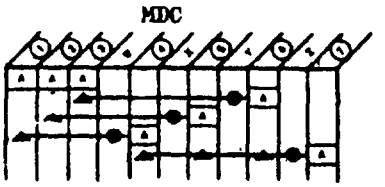
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f. Parallel relationships. Parallel relationships shall be shown as follows:

Schematic Block Diagram

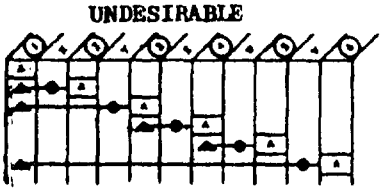


MDC

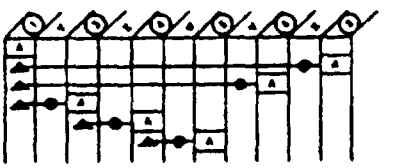


g. Serial parallel relationships. Serial parallel relationships shall be mapped as shown below as "CORRECT."

UNDESIRABLE

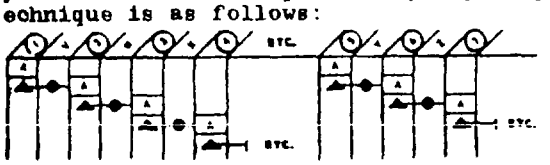


CORRECT



The undesirable presentation erroneously implies by its structural layout that there is one serial dependency chain. The correct presentation shows clearly by its structural layout that there are three separate parallel dependency chains. It also shows that one of the dependency chains has several relationships.

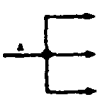
h. Horizontal line usage. Normally, only one dependency line shall be used per horizontal segment of the chart. More than one dependency line may be used, however, if necessary to fit all the required dependency chains on an MDC. In this case, dependency chains shall be separated by ample space to avoid confusion. An example of this technique is as follows:



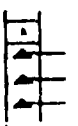
Notice that each dependency chain is easily identifiable.

i. Parallel divergent branches. The divergence of a single path into a number of paths shall be depicted by an event/dependency column using a single event and the appropriate number of dependency markers. This situation shall be depicted as shown in the following two illustrations.

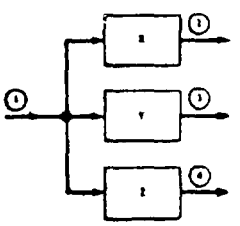
Schematic Block Diagram



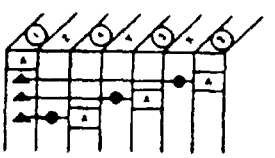
MDC



Schematic Block Diagram

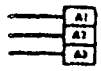


MDC



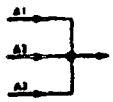
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j. Composite events. A composite event is made up of a number of single events. The constituent events may or may not be measured individually. A composite event shall be shown in an event/dependency column as follows:



k. Parallel convergent branches. When paths converge into a single path, they shall be depicted by an event/dependency column using a composite event and a single dependency marker. This situation shall be depicted as shown in the following two illustrations:

Illustration (1) Schematic Block Diagram



MDC

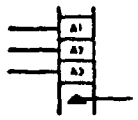
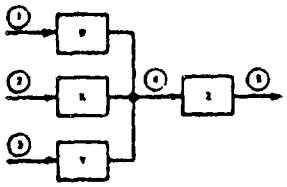
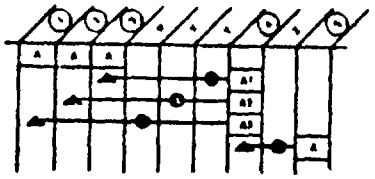


Illustration (2) Schematic Block Diagram

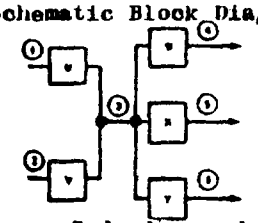


MDC

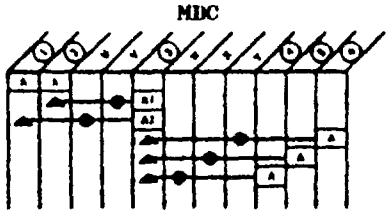


l. Convergent divergent branches. The following example illustrates a situation in which both divergent and convergent branches exist.

Schematic Block Diagram



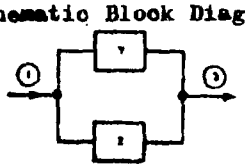
MDC



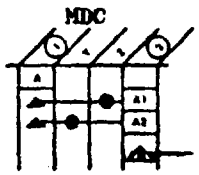
m. Redundancy. Redundancy shall be mapped using the composite event and a special dependency marker. The following examples illustrate the mapping of various types of redundancy.

(1) Active redundancy. Active redundancy shall be mapped as follows:

Schematic Block Diagram



MDC

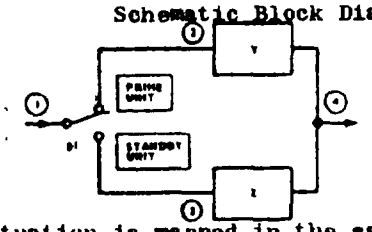


The MDC shows that if the signal at ① is available, and either component Y or Z is operating, the signal at ② will be available. The notation R within an open dependency marker denotes a redundant event.

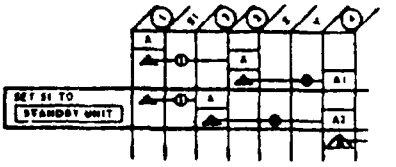
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(2) Manually switched redundancy. Manually switched redundancy shall be mapped as follows:

Schematic Block Diagram



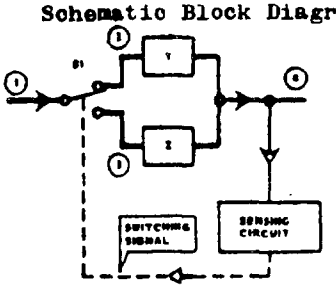
MDC



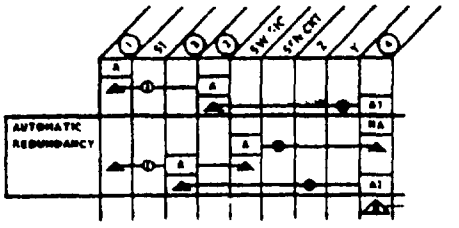
This situation is mapped in the same manner as the active redundancy situation except that the man operation is specified in the procedure column and outlined with a heavy line along with the associated dependency line.

(3) Automatically switched redundancy. Automatically switched redundancy shall be mapped as follows:

Schematic Block Diagram



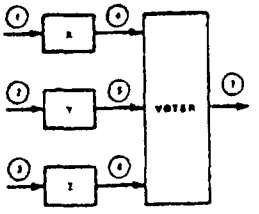
MDC



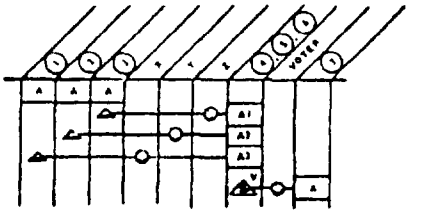
The MDC first maps the primary dependency structure. The automatic switching is emphasized by the heavy lines and the notation within the procedure column. The origin of the redundant signal shall be shown by the use of an availability box (3.2.8.3.2a) containing the symbol NA (shown at (4) in the illustration) which generates a switching signal to activate the redundant path.

n. Voting. The following example illustrates the mapping of a typical voting situation.

Schematic Block Diagram



MDC



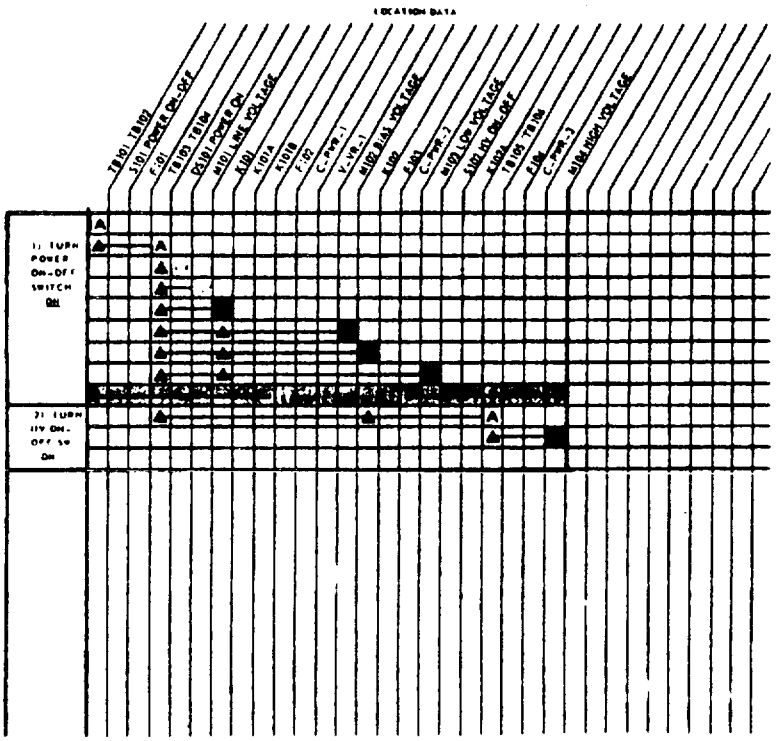
This example illustrates a 2/3 voter. Parallel events at (4), (5), (6) are shown as a composite event. The dependency marker under the composite event has special notation as follows: The V denotes that the events are being voted. The (2) within the dependency marker indicates that any two of the three events represented by the composite event are required for the event at (7) to occur.

o. Closed loop. For troubleshooting, a closed loop shall be isolated by heavy lines to encompass those units in the closed loop. This is to indicate to the technician that the loop must be opened for troubleshooting. See figures 17 and 18.

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3.2.7.6 Layout. The organization and layout of the MDC shown in Figure 9 shall be used. Procedural steps and their resulting dependencies shall be separated on the MDC with bold lines.

3.2.7.7 Depiction of time delays. Time delays after a procedure step shall be illustrated by placing a band of screen shading horizontally across the page immediately after the entry of the time delayed device. The time delay shall be assumed to start when the switch which activates it is thrown. The time delay length shall be indicated in the shaded time delay band. When a measureable time delay is inherent in an operation performed within a functional segment, it shall be shown as part of the signal specification which shall state when the time delay started and how long it will last.



3.2.8 Keyed text. The keyed text shall provide formatted information as to what the equipment in the schematic block diagram does. Phrases or paragraphs shall be keyed with reference letters to the equipment illustrated on the schematic block diagram. The keyed text shall contain statements as to principles of operation, use, etc. (See Figures 6, 6a, and 14.) The keyed text page shall precede the Schematic Block Diagram (SBD) to which it applies. It shall be placed on the upper page (back of the preceding page or start on an even numbered page) facing the SBD.

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3.2.8.1 Organization. The first paragraph in the keyed text shall be titled "General" and contain a general description on the system or subsystem. All components shall be identified in this paragraph. The operation of each component shall be described in a paragraph after the general paragraph. Each paragraph describing a component shall be identified with a block alphabetic character. Characters shall be assigned to the paragraphs in alphabetical order. The alphabetic character assigned to a paragraph describing a component shall also appear on the schematic block diagram adjacent the component.

3.2.8.2 Content. The keyed text paragraphs shall contain information relevant to the operation, use, and maintenance of the particular component, system or subsystem.

3.2.9 Schematic Block Diagram. Schematic block diagrams show in symbolic form the interrelationships of hydraulic, mechanical and electrical components within a system or subsystem. It shall consist of symbols that represent components which are identified on the locator illustration and are directly referenced to the locator illustration by callouts. Each component on a schematic block diagram shall be identified by common name and its location referenced to the locator page. (See Figures 7, 7a, and 15.) Schematic block diagrams define hardware boundaries, show complete interconnection information and use of special techniques and symbols to indicate signal flow. Layout is of prime importance and rigid rules are specified to insure proper layout and signal flow.

3.2.9.1 Funotional boundaries. Each component contained within a system or subsystem shall be indicated with a light solid line. (See Figure 7, 7a, and 15.)

3.2.9.2 Component name and callouts. Each component shall be depicted in a locator illustration and assigned a callout number. This callout and the component name (the verbal portion of the hardware name tag) shall be placed inside the component boundary. (See Figures 8 and 12.)

3.2.9.3 Layout. Components shall be arranged so that the normal signal or process flow will be from left to right. Feedback signal flow shall be from right to left. Input signals shall appear on far left of the page. All outputs shall appear on the far right of the page. Feedback signals to the system or subsystem shall appear on the far right of the page. (See Figures 7, 7a and 15.)

3.2.9.4 Signal lines shall not be grouped into one signal line. If the drawing including all the signal lines is too large for containment on a single 10 3/4 by 16½ page the drawing may be continued on as many pages as necessary to properly depict all relationships as specified in this document. (See figures 7 and 7a)

3.2.9.5 Signal names and waveforms. All input and output signals shall be given descriptive names. Signals generated with in the system, subsystem or component which are connected to output plugs, but which are not used shall be labeled "not used." Signals of importance which are generated within a component may be identified within the component by using a signal name flag pointing to the line to be identified. Waveforms and other signal specifications for test points shall be added for clarity.

3.2.9.6 Components which are physically located together will be grouped together on the drawing and the location identified as called out on the locator diagram.

3.2.9.7 Removable units and segments shall be identified by name and location as called out on the locator page (3.2.7).

3.2.9.8 Interconnecting circuitry such as electrical wiring, hydraulic plumbing, pneumatic ducting and mechanical linkages shall be included on the drawing. Terminal boards or other signal junction points where signal flow converges or diverges shall be shown.

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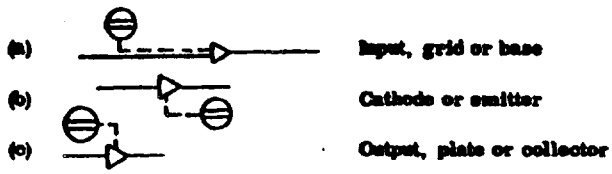
3.2.9.9 Connector identification numbers shall be shown on each side of the connector.

3.2.9.10 Test points. Designed test points, (connectors or other physical objects), shall be included when they occur between systems and subsystems or between components. Test points shall be shown schematically at their point of connection. Test point identification shall be as outlined in MIL-M-38800.

3.2.9.11 Adjustments and controls. Test points are illustrated the same as an adjustment. Adjustments and controls shall be illustrated to indicate the physical means of adjustment:

 Screwdriver,  Hexnut,  Knob, etc. The adjustment symbol and the adjustment nomenclature shall appear in the area representative of the hardware on which the adjustment is located.

3.2.9.11.1 The adjustment symbol shall be connected to the blocked symbol to indicate the relative circuit location of the adjustment as follows:



3.2.10 Supplemental Data Page (as required). The supplemental data page shall be used when required information cannot be adequately depicted in the other five basic page forms. Supplemental data may take the form of graphic material, tables, or symptom cause information. (See Figures 10 and 16.)

3.2.10.1 Organization. Supplemental data shall be organized in the best manner to suit the type material presented.

3.2.11 Alphabetical index. An alphabetical index shall be prepared for each SBD/MDC in accordance with MIL-M-38784. It shall be the last page of the SBD/MDC preceding the back cover.

3.2.12 Back cover. The back cover shall be blank unless the manual is classified, in which case the classification shall appear at the top and bottom center. When the back cover is blank, no reproducible copy is required, only the Reproduction Assembly Sheet need be noted that there will be a back cover.

4. QUALITY ASSURANCE PROVISIONS

4.1 The requirements of MIL-M-38784 and MIL-P-38790 are applicable.

5. PREPARATION FOR DELIVERY

5.1 The requirements of MIL-M-38784 and MIL-P-38790 are applicable.

6. NOTES

6.1 Intended use. Manuals prepared in accordance with the requirements of this specification are intended for use in the troubleshooting of various types of complex equipment.

6.2 Definitions. The following key terminology and symbols are used on maintenance dependency charts.

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6.2.1 Event. An action or an availability of a signal at a point in a circuit. The event may be characterized as motor running; temperature normal; lamps lit; lamps out; instructions indicating; signal or voltage available or not available; relay solenoids, or thrusters energized, etc.

6.2.2 Outlined rectangle. A rectangle with black lettering is a circuit point at which measurement might at some time be made. The circuit point is not necessarily readily accessible.

6.2.3 Functional entity. A group of circuit elements that together form a basic functional circuit, such as a filter network, a voltage divider network, an amplifier stage, an oscillator stage, a flip-flop stage, etc.

6.2.4 Black circle. Represents a circuit element or a functional entity.

6.2.5 Dependency marker. The solid black triangle is used as a dependency marker. The dependency marker is used on an event line to denote that an action or availability of a signal occuring on its line is dependent upon the occurence of an action or availability of a signal directly above its apex. The signal or action above the dependency marker must be available and within specification for the event on the line of the dependency marker to result, provided that all the circuits and parts symbolically represented along the line are also performing properly.

6.2.6 Part - One piece, or two or more pieces joined together which are not normally subject to disassembly without destruction of designed use.

6.2.7 Sub-assembly - Two or more parts which form a portion of an assembly or a unit replaceable as a whole, but having a part or parts which are individually replaceable.

Custodian:

Air Force - 16

Review Activities:

Air Force -01, 10, 70, 71, 80, 82, 84

Preparing Activity:

Air Force - 16

Project No. TMSS-F333

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TECHNICAL MANUAL

ORGANIZATIONAL MAINTENANCE

TROUBLESHOOTING AID

LANDING GEAR SYSTEM

F-5A

AIRCRAFT

SAMPLE

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JGM F-6A-2TS-8

The agency for mandatory use and responsibility for the Army is AR 310-1, for the Navy and Marine Corps is OPNAVINST 5510.1B, and for the Air Force is Air Force Regulation 205-1.

LIST OF EFFECTIVE PAGES

Insert latest changed pages, dispose of superseded pages in accordance with applicable regulations.

NOTE: On a changed page, the portion of the text affected by the latest change is indicated by a vertical line, or dot(s) adjacent to the outer margin of the page. Changes to illustrations are indicated by miniature pointing hands. Changed or added drawing data is indicated by shaded area.

Total number of pages in this manual is 14, consisting of the following:

Page No.	* Change No.
Title	0
A	0
i - ii	0
1 - 10	0

* Zero in this column indicates an original page.

A

Figure 2

TABLE OF CONTENTS

Section		Page
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1	LANDING GEAR SYSTEM	
	Preliminary Information	1
	Keyed Text	3
	Schematic Block Diagram	5
	Locator Page	6
	Maintenance Dependency Chart	7
	Supplemental Data Page	10

Figure 3

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This document consists of aids developed to assist in troubleshooting on-aircraft systems at the organizational maintenance level. Each troubleshooting aid consists of sections (1) Preliminary Information Sheets; (2) Keyed Text; (3) Schematic Block Diagrams; (4) Locator Page; (5) Maintenance Dependency Charts; (6) Supplemental Information Sheets.

PRELIMINARY INFORMATION SHEET

The Preliminary Information Sheet provides information on:

1. TEST EQUIPMENT which may be required,
2. PERSONNEL requirements, and
3. EQUIPMENT CONDITION.

The user must read and perform the instructions set forth on this sheet in order to configure the aircraft for the troubleshooting sequence contained in the troubleshooting aid. Failure to do so may result in personnel injury or equipment damage. In addition, this sheet may contain information relative to specific aircraft effectivity, such as the effect of TCTO on designated aircraft.

KEYED TEXT

The keyed text provides information as to what the equipment shown on the facing schematic block diagram does. Paragraphs are keyed with reference letters to the equipment illustrated on the schematic block diagram. They are also keyed to the locator page by a number following the letter designator. The text contains statements as to the principles of operation, peculiarities, etc.

SCHEMATIC BLOCK DIAGRAM

The schematic block diagrams show in symbolic form the interdependencies of hydraulic, electrical, mechanical, and pneumatic components within a system. They consist of symbols that represent components which appear on the facing locator page and are directly referenced to the locator page by callout numbers.

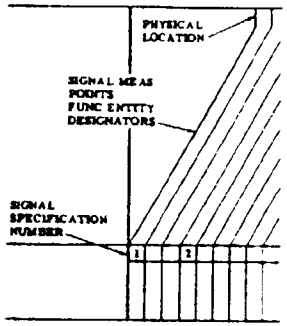
MAINTENANCE DEPENDENCY CHART

The Maintenance Dependency Chart (MDC) is a symbolic presentation of the dependency relationship between a group of functional entities and their related input and output signals or indications. The basic chart form consists of a heading, body, procedure column, and signal specifications.

HEADING

The chart heading lists the functional entities and their related input and output events. Each event is assigned a specification number used as a reference to the signal specifications contained in the table located at the right of each MDC. The heading also includes signal measurement points, functional entity designators, and physical location of the functional entities.

Signal measurement points and functional entity designators are located immediately above the signal specification number in the heading columns. Aircraft replaceable functional entities are indicated by the designator contained between vertical lines that do not extend to the heading border. Physical location information is at the extreme top of the chart heading between vertical lines extending to the heading border. Physical location information shall be in the form of a callout number from the facing locator page.



BODY

The Maintenance Dependency Chart body consists of a number of symbols connected by horizontal lines. The arrangement depicts the dependence of an indicated output upon the proper functioning of an entity and its required inputs. The symbols used in the chart body are divided into three categories

☐ EVENT

This symbol indicates input or output signals or indications. Additional entries within this symbol denote the nature of the event. The background of the event symbol denotes ease with which the event may be observed. Definitions of the entries used within the event symbol and the background meaning are explained in the definition of symbols table.

Each event symbol will have normal reading and tolerance information provided in the signal specification table. Proper signal specifications are referenced by numbers appearing at the top of the event columns. No signal specification will be provided for self evident events such as [redacted] and [redacted].

● FUNCTIONAL ENTITY

Variations to this symbol are shown in the definition of symbols table. The functional entity symbol represents system components which must be operating properly (proven good) before an event on the line can occur within specifications.

▲ DEPENDENCY MARKER

This symbol is used to show that the event on the same horizontal line is dependent upon the occurrence of the event directly above the triangle (in the same vertical column). The dependency marker shows the dependency relationship between different circuit paths (horizontal lines).

PROCEDURE

The procedure column identifies the steps necessary to effect the functional entity relationships identified directly to the right of the step in the body of the MDC. A sequential performance requirement is specified when dependency markers (▲) appear directly under availabilities [A] developed in preceding steps.

SIGNAL SPECIFICATIONS

The signal specifications column on the right side of the MDC contains information concerning the acceptable signal flow thru the system/subsystem. Each critical signal is keyed by a number appearing under the signal measurement point or subsystem interface column in the heading. Variation resulting from procedure step performance is denoted by subkeying. Signal specification numbers are developed in a left-to-right sequence, changes in signal strengths are developed in a top-to-bottom sequence.

MDC USE

When using the Maintenance Dependency Chart remember the chart is a symbolic representation of the signal flow thru the equipment. Thus, as in the case of a diagram, the input event signals are shown on the left side of the chart body and output event signals toward the right side of the chart body.

As a troubleshooting tool, the chart should be entered at the faulty event under the same mode or condition of equipment as indicated by the procedure column of the chart. This event is the first bad indication.

Next the technician attempts to bracket the failure by locating a good event within the dependency structure associated with the faulty event. This good event should be the last good indication.

During the bracketing procedure, the technician should make maximum use of all front panel and easily accessible indications.

With knowledge of the 'last good indication' and thru the use of front panel and easily access ble indications, the technician will also be able to verify certain events and functional entity associated with the 'first bad indication.' (All troubleshooting conclusions may be marked on the chart to maintain an awareness of checks performed and alternatives remaining.)

Referring to the dependency line of the 'first bad indication,' the technician investigates the dependency symbols on this line, first checking the dependency markers (▲) to determine if the events above them have occurred and are within specification.

If all such events are good, the malfunction lies on the line of the first bad indication and the functional entities (●) on this line are checked.

If one of the events above a dependency marker is proved bad, the technician immediately applies his attention to the dependency line associated with that event and repeats the above procedure.

This process is repeated until the faulty functional entity is located.

INTRODUCTION

Figure 4

DEFINITION OF SYMBOLS

CATEGORY	SYMBOL	MEANING
DEPENDENCY MARKER	▲	Event on same line depends upon event or condition in column under marker.
	△	Event depends upon a redundant event or condition in column above marker.
EVENT	■	Event symbol background indicates ease of access. Black: Front panel, recognized from outside the equipment.
	□	Screen: Easy access, requires opening of equipment enclosure (may or may not require test equipment).
	□	White: Difficult access, requires test equipment.
	[A]	Signal not available.
	[▲]	Signal available and within specification.
	[▲1] [▲2] etc	Signal available and within specification, but under different procedural condition or circumstance.
	[L]	Front panel indicator lit, location at...
	[F]	Indicator flashing.
	[F]	Motor runs within specification.
	[R]	Relay or special purpose switch energized.
	[R]	Relay or special purpose switch deenergized.
	[F]	Motor stops running.
	[F]	Previously lit indicator goes out.
	[C]	Functional entity closed.
	[O]	Functional entity open.
	[E]	Functional entity extended.
	[R]	Functional entity retracted.
FUNCTIONAL ENTITY	●	Circuit or circuit element that must function properly for event on same line to occur.
	⊙	Indicates functional entity requires more than one event to prove proper operation.
	⊣⊢	Relay open.
	⊣#⊢	Relay closed.
	⊣ ⊢	Switch open.
	⊣#⊢	Switch closed.
	⊖	Valve closed.
	⊖	Valve open.
	●●	Redundant circuit elements: either will produce event. (May be used with switches, relays, valves, or other circuit elements.)

F-1a-TTS-4-1

LANDING GEAR SYSTEM
PRELIMINARY INFORMATION

APPLICABLE SERIAL NUMBERS

▲ AF66-9129 thru AF66-9151, AF66-9139 and AF66-9140 aircraft only.

▲ AF65-10408 and earlier aircraft not modified by T.O. 1F-5-624.

▲ AF65-10500 and later and aircraft modified by T.O. 1F-5-624

▲ AF64-13910 and later aircraft.

TEST EQUIPMENT

Table X-XX-X. Test Equipment Required

NAME/ TYPE	FEDERAL STOCK NUMBER	USE AND APPLICATION
GENERATOR SET, 400-cycle, 3-phase AFM32A10 or MILM4617 TYPEMD1	6115-617-1940 6115-988-6760	Provide source of external electrical power.
STAND, HY-draulic Test MILS28877 TYPEMU3A or MILS28017 TYPEMK3A	4920-618-8883 4920-529-8780	Provide source of external hydraulic pressure.
MULTIMETER AN/PSM6B PANEM	6625-726-6582	Measure voltage and continuity in landing gear system.
TESTER, Pitot-Static System MILT7427 TYPETTU9E or MB-1A	4920-613-2987	Test landing gear warning and indication system.
JACK, Hy-draulic Hand MILJ26913 TYPE A3	1730-560-2343	Jack aircraft at landing gear.
ADAPTER, Jack 8-76877	1730-715-4521	Provide jack pad for wing and fuselage jacking points.

PERSONNEL REQUIRED

Personnel Required: Four

Four men, two with intercommunication sets, are required to perform landing gear operational checks. Station man A is the cockpit to operate landing gear controls. Station man B is required to observe movement of landing gear, to communicate with man A, and to direct operations of men C and D. Station men C and D to observe landing gear operations and make necessary measurements and alignment checks.

A qualified person will be required to operate electrical power unit.

A qualified person will be required to operate hydraulic test stand.

Instrument specialist will be required to operate pitot-static tester.

NOTE

If multiple-place interphone ground stations are required, an external power supply is necessary. Refer to T.O. XXX for additional information.

EQUIPMENT CONDITION

Before checking landing gear operation, check the following conditions:

WARNING

- Do not connect electrical power or apply hydraulic pressure to aircraft unless specifically instructed to do so.
- All external stores must be disarmed and removed and guns cleared as described in T.O. XXX.

CAUTION

Landing gear struts will be serviced in accordance with applicable instructions. Struts, tires, and airframe structure will be damaged if struts are not fully extended before landing gears are retracted.

1. Service main and nose gear struts.
2. Jack aircraft in accordance with applicable instructions. Use forward fuselage and wing jacking points.
3. Remove landing gear ground safety pins (three placed).
4. Check that nosewheel steering disconnect pin (9) is properly engaged.
5. Check that gear doors control switch (6) in right main gear wheel well is at NORMAL.
6. Have external electrical power source available.
7. Connect hydraulic test stand to utility hydraulic system quick-disconnects.
8. Check that ejection seat and canopy ground safety pins are installed in accordance with applicable instructions.

CAUTION

Failure to open left and right engine fuel circuit breakers may result in engine fire when throttle controls are advanced during procedure.

9. Open (pull out) the following circuit breakers on the pedestal dc circuit breaker assembly:
   a. RELEASE & BOMB ARM CONT (3)
   b. L ENG START & AB CONT (4)
   c. R ENG START & AB CONT (5)
10. Close (push in) the following circuit breakers on the dc power distribution box in the right forward fuselage section:
    a. LANDING GEAR CONT, SAFETY & STEERING (13)
    b. LANDING GEAR WARNING (12)
    c. CAUTION & WARNING LIGHTS-BRIGHT (11)
    d. AN AIC-10 INTERPHONE (10)
11. Check that left and right throttle control levers are at OFF.
12. Check that left and right fuel shutoff valve control switches (1) are at CLOSED.
13. Check that landing gear control handle (14) is at LG DOWN.
14. Connect ground crew headset-microphone intercom service cord to the interphone ground station receptacle (8).
15. Electrically ground aircraft in accordance with applicable instructions.

WARNING

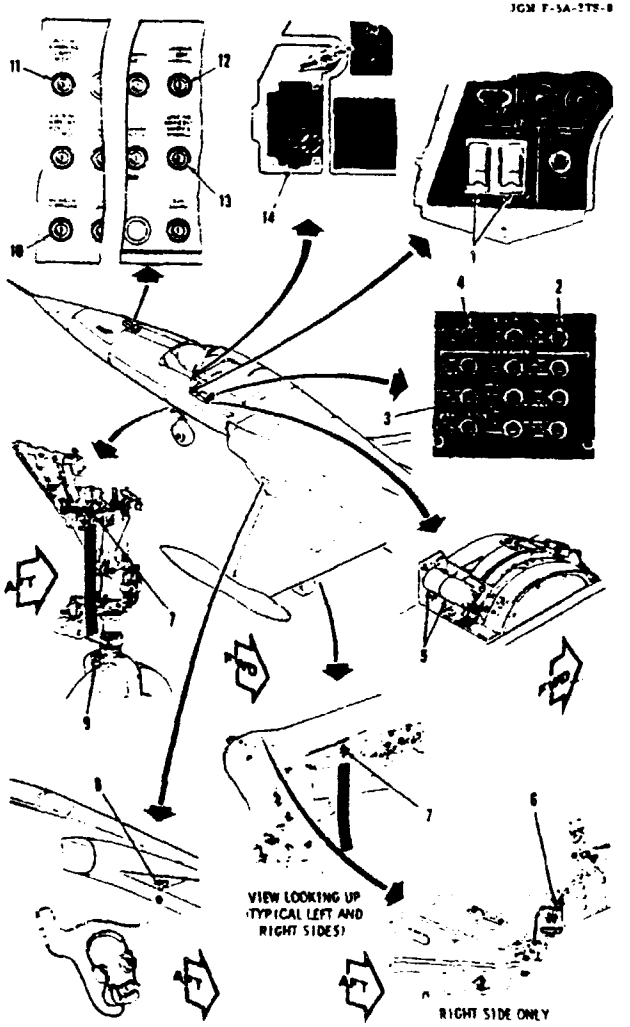
- Be sure personnel and equipment are clear of landing gear doors when actuating gear doors control switch. Doors may close or open when electrical power and hydraulic pressure are applied if switch was actuated while power was off.
- Landing gear doors may open inadvertently during ground operations whenever all the following conditions are met: Engines are not operating, electrical power is applied to aircraft, utility hydraulic pressure is applied, throttle levers are below the 90° stop setting, dc circuit breaker marked LANDING GEAR WARNING is open, dc circuit breaker marked LANDING GEAR CONT, SAFETY & STEERING is closed, landing gear control handle is at LG DOWN and warning test switch on gear control panel is held at TEST position. Release of test switch will close doors. When this condition exists, a circuit is completed to ground thru the landing gear door sequence relay.

CAUTION

- During operational procedure, gear action may be halted immediately by opening bypass valve on hydraulic test stand.
- To prevent possible damage to nose gear, Alternate release linkage, do not pull Alternate release handle to open gear doors with nose gear in extended position.
- While landing gears are slowly retracting, check for adequate clearance between gears and adjacent electrical wiring and switches, hydraulic lines and fittings, door and switch linkage, control cables, and gear doors.
16. Apply electrical power to aircraft in accordance with applicable instructions.
17. Apply hydraulic pressure to aircraft in accordance with applicable instructions.

NOTE

- Adjust hydraulic test stand to deliver 2 to 10 gal min at 3000 lbf in^2.
- Landing gears should retract to uplocked position with hydraulic pressure of not more than 2700 lbf in^2.



INPUT CONDITIONS PAGE - LANDING GEAR SYSTEM

Figure 9

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LANDING GEAR SYSTEM FUNCTIONAL DESCRIPTION.

GENERAL. The landing gear system includes a gear control panel in the lower left corner of the main instrument panel, an alternate release cable system operated by a handle on the left vertical control panel, hydraulically actuated main and nose landing gear, hydraulically actuated main gear inboard doors and nose gear forward door, and strut fairing doors mechanically linked to the struts. Each main gear has a hydraulically actuated uplock and a mechanical overcenter downlock. Nose gear uplock and downlock assemblies are the mechanical overcenter type. Each main gear inboard door and the nose gear forward door actuate door-open sequence switches, which control sequencing of electrical power to energize the landing gear and landing gear doors hydraulic selector valves. Thus, landing gears are allowed to move only after the gear doors have fully opened. The electrically controlled landing gear and gear door selector valves direct hydraulic pressure to the gear and gear door actuating cylinders and uplock actuators. Main gear inboard doors and the nose gear forward door may be opened or closed while the aircraft is on the ground by actuating a switch in the right main gear wheel well. The switch, placarded GEAR DOORS (NORMAL and OPEN), will operate the doors if hydraulic pressure and electrical power are available and the gear control handle is at LG DOWN. The gear doors switch should be placed at OPEN only for ground maintenance or inspection of the landing gear and should remain in the NORMAL position at all other times. Electrical power to operate the landing gear control system and landing gear safety system is furnished by the aircraft 28-volt dc system. Circuit protection is provided by a 5-ampere circuit breaker placarded LANDING GEAR CONT, SAFETY & STEERING in the nose equipment compartment dc power distribution box. The landing gear warning control functions on 28-volt dc and is protected by a circuit breaker placarded LANDING GEAR WARNING. The landing gear warning circuit controls dc power to the audible warning signal generator and landing gear warning control relays, which provide a ground for landing gear system warning lights. Electrical power for gear position green indicator lights and gear control handle red warning light is 28-volt dc (bright) or 14-volt dc (dim), operating thru the caution, warning, and indicator system dimming and testing controls. All control relays for landing gears, gear doors, and landing gear warning are on electrical relay assembly No. 1.

LANDING GEAR CONTROL PANEL. The landing gear control panel consists of a gear control handle, three green-in-top green indicator lights, a button placarded DOWNLOCK OVERRIDE, and a switch placarded WARNING SILENCE. The gear position indicator lights are lighted only when their respective gears are down and locked. The gear control handle red warning light is lighted whenever landing gears are not up and locked or down and locked, or whenever either main gear inboard door or the nose gear forward door is out of position (not closed and locked). The gear control handle red warning light will also come on whenever landing gears are down and locked and the gear doors control switch is right main gear wheel well is at OPEN. A landing gear audible warning signal is heard thru the headset and alerts the pilot to any abnormal condition in the landing gear system. The warning signal may be silenced by pressing the switch placarded WARNING SILENCE. To prevent inadvertent retraction of the landing gears, a solenoid-operated stop prevents the gear control handle being moved from LG DOWN position when aircraft weight is on the struts. The safety stop can be overridden by pressing the button placarded DOWNLOCK OVERRIDE.

LANDING GEAR WARNING LIGHT AND AUDIBLE SIGNAL CIRCUIT. The landing gear warning light and audible signal circuit visually and aurally indicates to the pilot whether the landing gears are in the down and locked position during landing configuration. The landing gear control handle warning light comes on whenever the landing gears are not up and locked (landing gear control handle at LG UP) or not down and locked (landing gear control handle at LG DOWN) or whenever either main gear inboard door or the nose gear forward door is not closed and locked (landing gear control handle at LG UP). The landing gear control handle warning light and the landing gear warning signal generator are energized whenever the landing gears are up and locked (landing gear control handle at LG UP) and the following conditions are met: Airspeed is below 220 ± 15 knots, altitude is below 10,000 ± 500 feet and either left or right throttle lever is set at less than 94% throttle setting. With the airspeed pressure switch, altitude pressure switch, and a throttle position switch closed, if one of the landing gear downlock switches is not in the down and locked positions, ground return is provided thru the switch to energize the landing gear warning signal and light relay. Also, with one of the downlock switches not in the down and locked position, ground return is provided thru the landing gear control handle switch (landing gear control handle at LG DOWN) to energize the landing gear warning light relay. When either relay (landing gear warning light or landing gear warning signal and light) is energized, ground return is provided for the illumination of the landing gear control handle warning light and dc power from the LANDING GEAR WARNING circuit breaker is supplied thru the energized landing gear warning signal and light relay and the deenergized landing gear warning signal silence relay to the landing gear warning signal generator. The signal generator provides an audible signal to the pilot to inform him that all landing gear downlock switches are not in the down and locked position. The signal generator may be silenced by pressing the gear warning signal silence switch on the landing gear control panel. Pressing the silence switch energizes the landing gear warning signal silence relay, which removes the dc power input to the signal generator. When the airspeed pressure switch or altitude pressure switch is open or both throttle position levers are set at a throttle setting of 94% or greater and one of the landing gear downlock switches is not in the down and locked position, ground return is provided thru the landing gear control handle switch (handle in LG DOWN position) to energize the landing gear warning light relay. The energized landing gear warning light relay provides ground return to illuminate the landing gear control handle warning light. During this condition, the landing gear warning signal generator will not come on. The landing gear control handle warning light, the gear position indicator lights, and the landing gear warning signal generator can be tested by the warning test switch on the lighting control panel. The warning test switch held at TEST provides a ground return to illuminate the lights and allows dc power from the LANDING GEAR WARNING circuit breaker to be supplied to the landing gear warning signal generator thru the deenergized landing gear warning signal silence relay. Another warning has been provided in the landing gear warning light and audible signal circuit. This circuit is actuated by the landing gear doors control switch in the right main wheel well. The circuit is actuated when the landing gear doors control switch is placed in the OPEN position. This switch must be manually set into position. When the switch is placed at OPEN, ground return is provided thru either throttle position switch, the altitude pressure switch, and the airspeed pressure switch to energize the landing gear warning signal and light relay. The energized condition of the landing gear warning signal and light relay illuminates the landing gear control handle warning light and energizes the landing gear warning signal generator, providing the pilot with an audible signal and warning light indication that the landing gear doors control switch is not in the proper position. Before flight, the landing gear doors control switch must be placed at NORMAL position.

NOTE

Refer to Sheet 2 for functional description of landing gear retraction and extension.

KEYED TEXT

A The landing gear control panel consists of a gear control handle, five single-pole double-throw switches actuated by the control handle, three gear position indicator lights, two red warning lights, a downlock override button, and a warning silence switch.

B The three green gear position indicator lights (nose, left main, and right main gear) are lighted when all gears are down and locked.

C The control handle red warning lights are lighted whenever the gears are not up or down and locked or whenever the gears are up and locked and the doors are open.

D When the gear warning silence switch is pressed, the landing gear warning signal silence relay is energized, removing the dc power input to the warning signal generator.

E The downlock override button, when pressed, releases a solenoid-operated stop, which prevents the gear control handle being moved from the LG DOWN position when aircraft weight is on the struts.

F The alternate release reset switches disconnect electrical power to the landing gear selector valves and sequencing system.

G The LANDING GEAR CONT, SAFETY & STEERING circuit breaker supplies dc power to the normal landing gear control circuits and nosewheel steering circuit.

H The LANDING GEAR WARNING circuit breaker supplies dc power to the landing gear warning audible signal circuit and landing gear warning control circuit.

I The gear doors control switch will operate the doors if hydraulic pressure and electrical power are available and the gear control handle is in the LG DOWN position.

J When pressed, the warning test switch will test the landing gear handle warning lights, the gear position indicator lights, and the signal generator.

K The airspeed-altitude pressure switch electrical contacts close when the airspeed is below 220 ±15 knots and the altitude is below 10,000 ± 500 feet.

L The left and right throttle position switch electrical contacts close when the respective throttle lever is set below the 94% engine rpm position.

M The light dimming resistor assembly contains resistors and associated interconnecting elements. Three of the resistors decrease light intensity of the gear position indicator lights.

N The electrical relay assembly No. 1 contains the landing gear control and warning relays.

O The landing gear warning light relay is energized whenever the landing gears are not up and locked or down and locked or whenever either main gear inboard door or the nose gear forward door is not closed and locked (landing gear control handle at LG UP), the red warning lights will then be lighted.

P When the landing gear warning signal silence relay is energized by pressing the gear warning silence switch, the dc power input to the signal generator is removed.

Q The nose gear downlock indicator relay is energized whenever the nose gear is in the down and locked position. When the relay is energized, electrical power is directed to the gear selector valve gear extend solenoid.

R The landing gear door sequence relay is energized by placing landing gear control handle in LG UP position when gears are down and locked or by placing control handle in LG DOWN position when gears are up and locked. When the relay is energized, electrical power is directed to the gear doors selector valve door open solenoid. The relay may also be energized by placing the gear doors control switch at OPEN when the gears are in the down and locked position.

S The gear retract control relay is energized in all gear and gear door positions except gears up and locked and doors closed and locked. When the relay is energized, electrical power may be directed to the gear selector valve retract solenoid, and door selector valve open solenoid. An additional set of contacts provides a ground circuit for the gear warning light relay.

T The landing gear warning signal and light relay is energized when the airspeed is below 205 knots, altitude below 9500 feet, either left or right throttle lever is set below 88% rpm position, and any one of the landing gears is not down and locked. The audible signal will then be energized and the red warning lights will be lighted.

U The warning signal generator provides an audible signal, heard thru the headset, to inform the pilot that all landing gear downlock switches are not in the down and locked position.

F-5A-2TS-8-3

KEYED TEXT -- LANDING GEAR SYSTEM (SHEET 1 OF 2)

Figure 4

LANDING GEAR RETRACTION. With all landing gears down and locked and main gear inboard doors and nose gear forward door closed and locked, the three green indicator lights on the gear control panel are lighted to indicate a normal condition, the gear control handle red warning light is out, and the audible warning signal is off. With the gear struts extended (weight of aircraft off the wheels), a ground return is provided thru the nose gear downlock switch to energize the nose gear downlock indicator relay, allowing dc power from the LANDING GEAR CONT, SAFETY & STEERING circuit breaker to be supplied to the landing gear selector valve extend solenoid, thru the nose gear downlock indicator relay and gear control handle switch. The energized selector valve allows hydraulic pressure to be applied to the "down" side of the landing gear actuating cylinders. At the same time, dc power from the LANDING GEAR CONT, SAFETY & STEERING circuit breaker energizes the gear retract control relay; a ground return is provided thru the alternate release switches (No. 2), a gear control handle switch, and the gear uplock switches. Moving the gear control handle to LG UP energizes the gear doors sequence relay by providing a ground return thru a gear control switch and the gear uplock switches for the dc power supplied thru the LANDING GEAR CONT, SAFETY & STEERING circuit breaker. Energizing the gear doors sequence relay allows dc power to be supplied to the gear doors selector valve open solenoid, which directs hydraulic pressure to the main gear inboard door locks, main gear inboard door actuating cylinders, and the nose gear forward door actuating cylinder. The main gear inboard door lock hooks open, and the main gear inboard and nose gear forward doors open, actuating door-open sequence switches. When all three door-open sequence switches are actuated, the gear selector valve retract solenoid is energized by a ground return thru a gear control handle switch, the door-open sequence switches, and the gear retract control relay. The energized gear selector valve directs hydraulic pressure to the "up" side of the main and nose gear actuating cylinders, main gear sidebrace torque actuating cylinders, and main gear uplock actuators. Each sidebrace torque actuating cylinder applies a rotational force to the sidebrace assembly to assist the main gear actuating cylinder in retracting the main gear. As the main gear strut roller rotates the uplock hook past center, hydraulic pressure in the uplock actuator is applied to the uplock, forcing it into the locked position. When all landing gear are up and locked and uplock switches are actuated, the gear doors sequence relay is deenergized, which deenergizes the gear doors selector valve open solenoid, removing hydraulic pressure from the gear doors "open" side of the actuating cylinders. With the gear doors sequence relay deenergized, a ground return for the gear door selector valve close solenoid is provided thru the sequence relay closed contact and gear door locks switches. Hydraulic pressure, directed thru the gear door selector valve, will then close and lock the main gear inboard doors and nose gear forward door. Initial closing motion of either main gear inboard door or the nose gear forward door will deenergize the gear selector valve retract solenoid and shut off hydraulic pressure to gear actuating cylinders and main gear uplock actuators. When the main gear inboard doors and nose gear forward door are closed and locked, the gear door locks switches actuate, (switch open) removing dc power from the gear selector valve close solenoid, shutting off hydraulic pressure to the gear door actuating cylinders and main gear inboard door lock actuators. When the gear control handle was moved to the LG UP position, the control handle red warning light came on; as each gear unlocked from its downlocked position, its respective green indicator went out and the warning signal generator was energized. When all gears reached the up and locked position and the gear doors reached the closed and locked position, the control handle red warning light went out and the warning signal generator deenergized.

LANDING GEAR EXTENSION. With the landing gear up and locked and gear doors closed and locked, the three green indicator lights on the gear control panel are out, and the gear control handle red warning light is out. Moving the gear control handle to LG DOWN provides a ground return thru the alternate release reset switch (No. 2), a gear control handle switch, and the gear downlock switches to energize the gear doors sequence relay. This will direct dc power thru a sequence relay contact to energize the gear doors selector valve open solenoid. When the control handle was moved to LG DOWN, a ground return was also provided thru the alternate release switches (No. 2), a control handle switch, and the gear uplock switches to energize the gear retract control relay. Energizing the gear doors selector valve directs hydraulic pressure to the main gear inboard door locks, main gear inboard door actuating cylinders, and the nose gear forward door actuating cylinder. The main gear inboard door lock hooks will open, and the main gear inboard doors and nose gear forward door will open. As each gear door reaches the fully open position, it actuates a gear door-open sequence switch. When all three gear door-open sequence switches are actuated, a ground return is provided thru the sequence switches and a gear control handle switch to energize the gear selector valve extend solenoid. The energized selector valve allows hydraulic pressure to be applied to the "open" side of the main gear uplock actuators and the "down" side of the landing gear actuating cylinders, the main gear uplocks open, and the landing gears extend, being assisted into a downlocked overcenter position by spring tension pressure. The rate at which the landing gears extend is controlled by restrictors in the gear-down hydraulic lines. When all the gears are in the downlocked position and the downlock switches are actuated, the gear doors sequence relay will deenergize, and the nose gear downlock indicator relay will energize. With the sequence relay deenergized, a ground return for the gear doors selector valve close solenoid is provided thru a sequence relay closed contact and gear door lock switches. Hydraulic pressure directed thru the gear door selector valve will then close and lock the main gear inboard doors and the nose gear forward door. When all the gear doors are closed and locked, the gear door lock switches actuate (contacts open), removing dc power from the gear doors selector valve close solenoid, shutting off hydraulic pressure to the gear door actuating cylinders and main gear inboard door lock actuators. Initial closing motion of the main gear inboard doors and the nose gear forward door will actuate (open) each gear door sequence switch. However, dc power supplied thru the closed contact of the previously energized nose gear downlock indicator relay acts as a holding circuit to keep the gear selector valve extend solenoid energized. Hydraulic pressure will continue to pass thru the selector valve to the down side of the landing gear actuating cylinders. When the gear control handle was moved to LG DOWN, the gear control handle red warning light came on, the warning signal generator was energized, and as each landing gear downlock switch was actuated (closed), its respective green indicator light came on. When all landing gears reached the down and locked position, the control handle red warning light went out, and the warning signal generator was deenergized.

KEYED TEXT

V The gear doors selector valve, electrically controlled, directs hydraulic pressure to the gear door actuating cylinders.

W The gear selector valve, electrically controlled, directs hydraulic pressure to the main and nose gear actuating cylinders.

X The nose gear door actuating cylinder opens and closes the nose gear forward door.

Y When all three gear door open sequence switches are actuated (doors open), the gear selector valve is electrically positioned to direct hydraulic pressure to the retract or extend cycle.

Z The main gear inboard door lock switches and the nose gear door closed switch provide a ground for the gear retract control relay coil when the gears are up and locked and the gear doors are open. The switches also provide an electrical circuit to the gear door selector valve close solenoid when the doors are open.

AA The main gear inboard door locks (two locks per door) secure the doors in the closed position.

AB The main gear inboard door actuating cylinders open and close the inboard doors.

AC The nose gear strut, hydraulically retracted, retracts by hinging forward. A separate actuator provides centering steering.

AD The main gear struts, hydraulically retracted, are identical in operation but opposite in direction of retraction.

AE The nose and main gear actuating cylinders retract and extend the gears.

AF The main gear torque actuating cylinders apply a rotational force to the sidebrace assembly to assist the main gear actuating cylinders during the retraction cycle.

AG The main gear uplock actuators open and close the main gear uplocks.

AH The main gear and nose gear downlock switches provide a ground for the gear position indicator lights when the gears are down and locked; the switches also provide a ground for the gear doors sequence relay coil when the gear is extending. An additional circuit on the nose gear switch provides a ground for the nose gear downlock indicator relay coil when the nose gear is down and locked.

AI The main gears and nose gear uplock switches provide a ground for the gear retract control relay coil when the gear is not in the up and locked position. They also provide a ground for the gear doors sequence relay coil when the gear is retracting.

AJ The right main gear outboard torque link switch provides an electrical circuit to release a solenoid-operated stop in the gear control panel when the gear strut is extended.

AK The one-way hydraulic restrictors (5 places) control the rate at which the gears extend into the down and locked position and main gear inboard doors close.

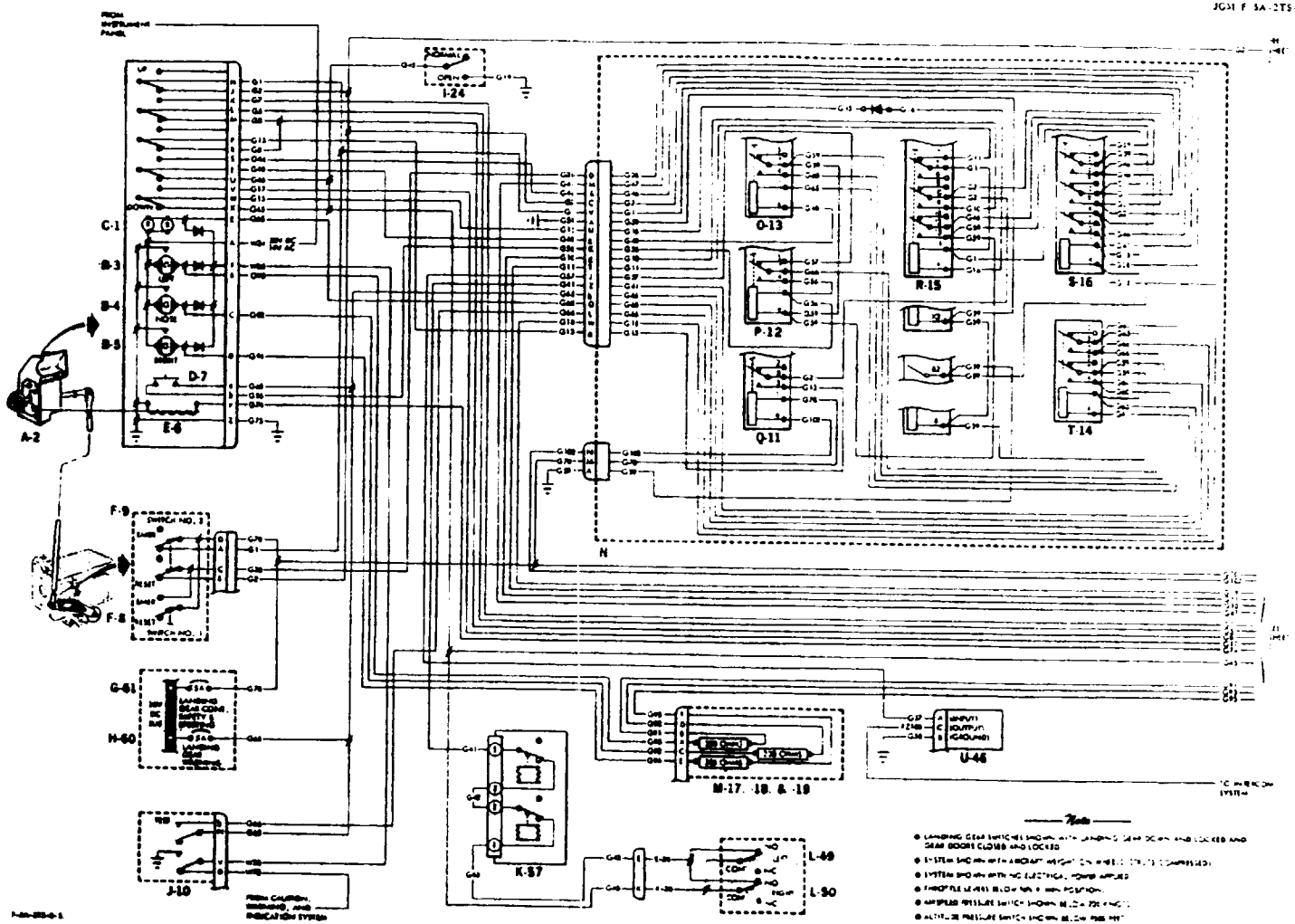
AL The two-way hydraulic restrictors (11 places) control the rate at which the gear doors open and close and prevent door-chatter during the door close cycle.

F-5A-2TS-8-4

KEYED TEXT - LANDING GEAR SYSTEM (SHEET 1 OF 2)

Figure 4a

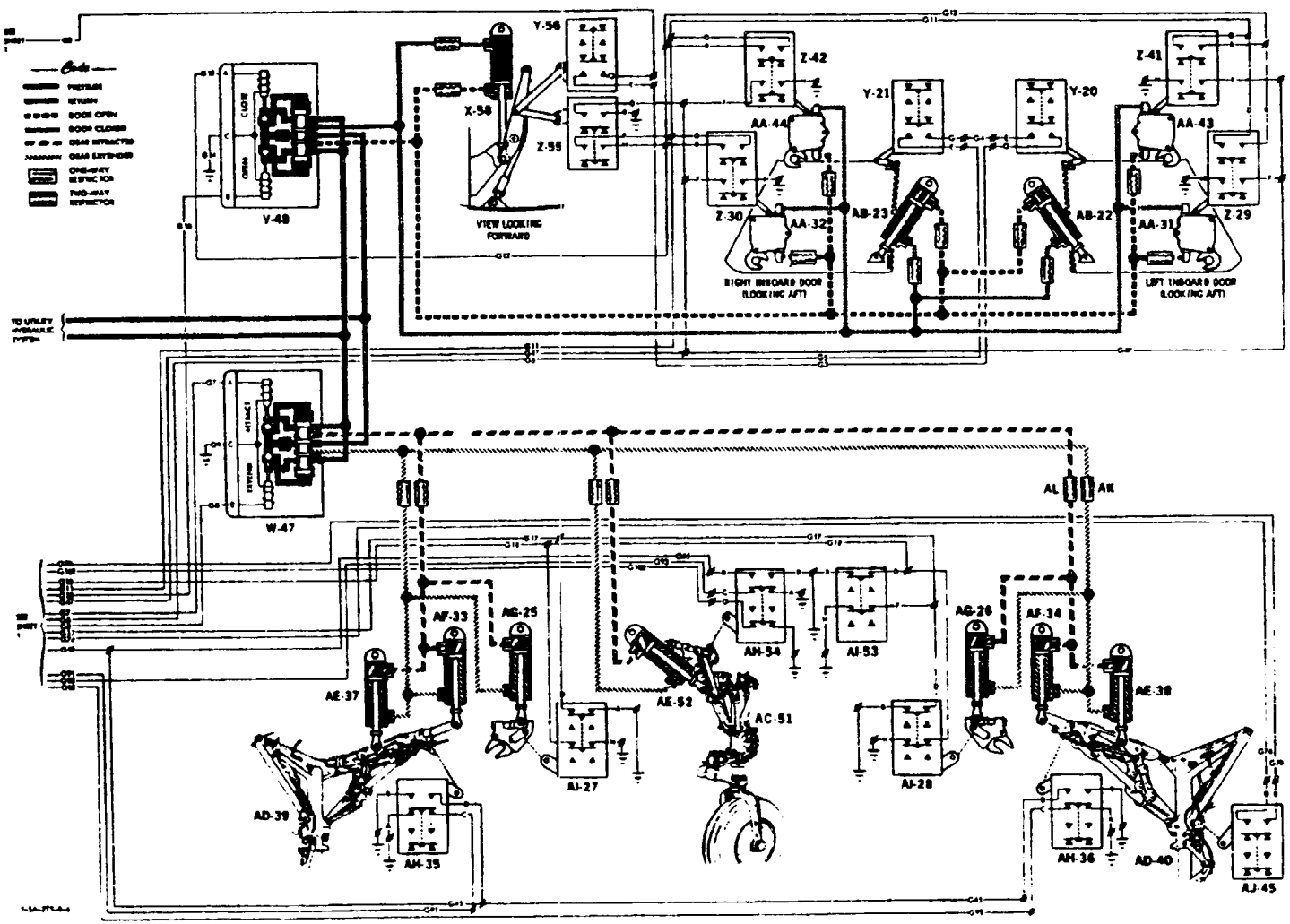
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SCHEMATIC BLOCK DIAGRAM - LANDING GEAR SYSTEM (SHEET 1 OF 2)

Figure 7

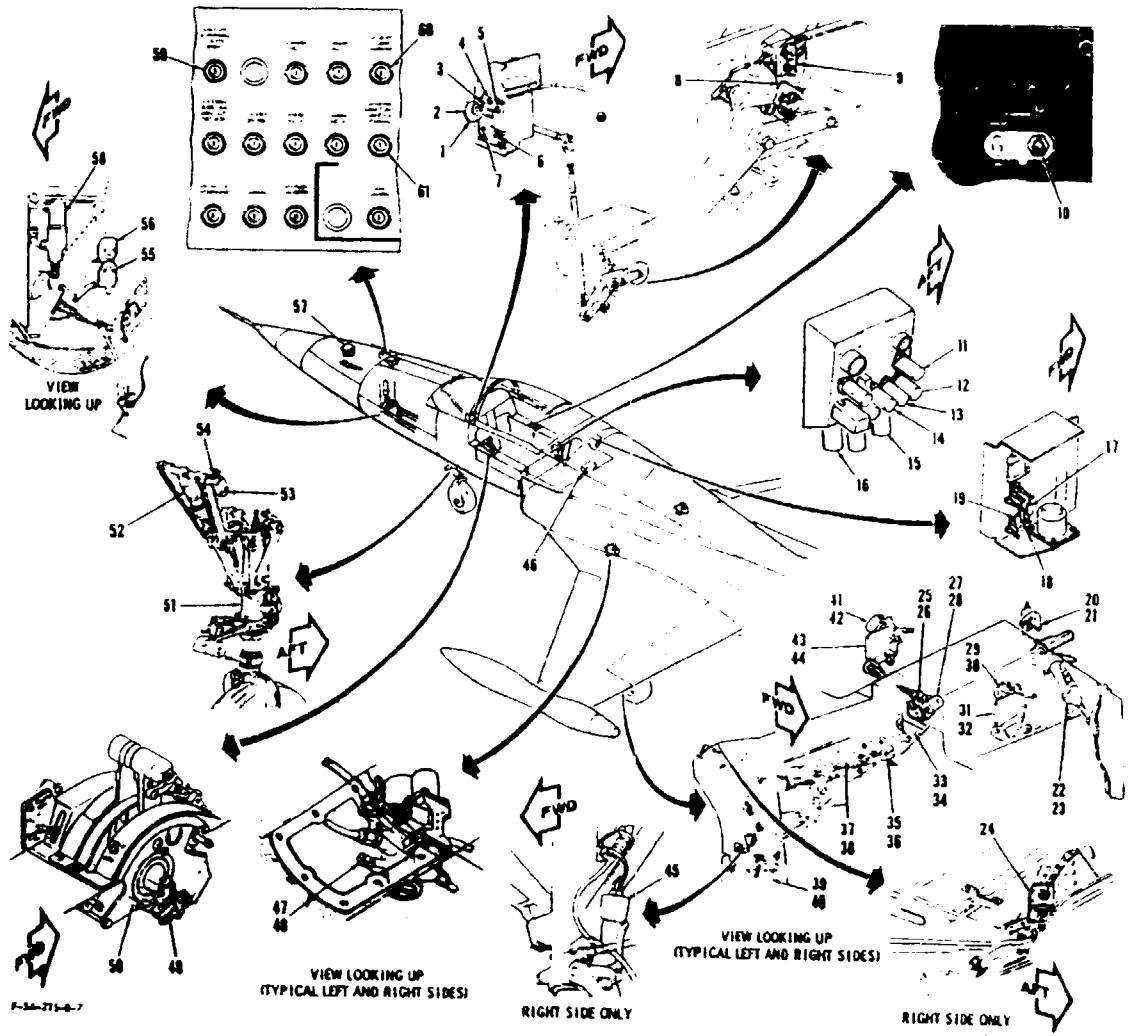
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SCHEMATIC BLOCK DIAGRAM - LANDING GEAR SYSTEM (SHEET 2 OF 2)

Figure 7a

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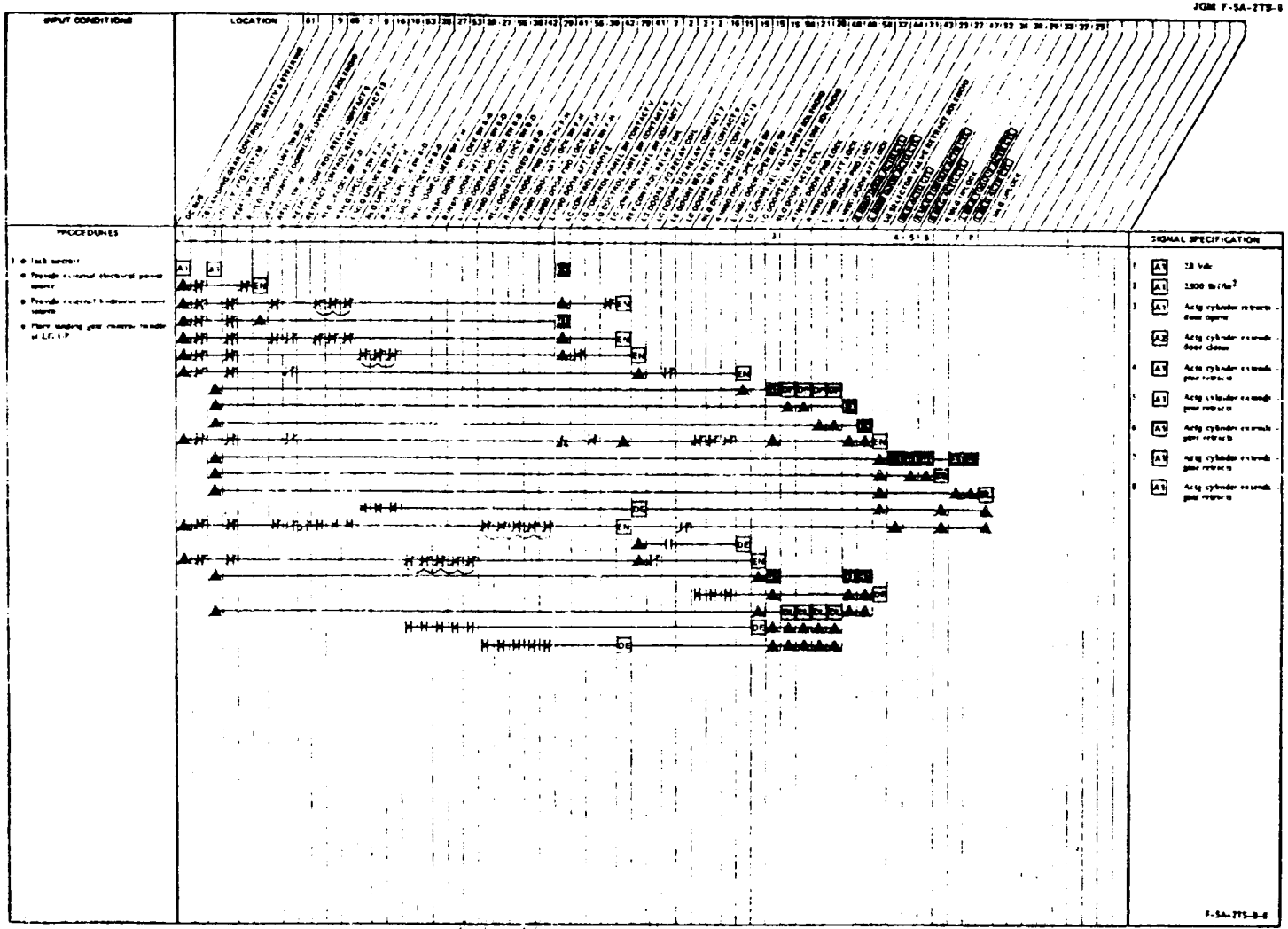
LOCATOR – LANDING GEAR SYSTEM

FIGURE 9

JGM F-5A-2TS-6

1 GEAR CONTROL HANDLE WARNING LIGHT
2 LANDING GEAR CONTROL HANDLE
3 LEFT GEAR POSITION LIGHT
4 NOSE GEAR POSITION LIGHT
5 RIGHT GEAR POSITION LIGHT
6 LANDING GEAR HANDLE DOWNLOCK OVERRIDE BUTTON
7 GEAR WARNING SILENCE SWITCH
8 RESET SWITCH NO. 1
9 RESET SWITCH NO. 2
10 WARNING TEST SWITCH
11 NOSE GEAR DOWNLOCK INDICATOR RELAY
12 GEAR WARNING SIGNAL SILENCE RELAY
13 GEAR WARNING LIGHT RELAY
14 GEAR WARNING SIGNAL AND LIGHT RELAY
15 GEAR DOOR SEQUENCE RELAY
16 GEAR RETRACT CONTROL RELAY
17 WARNING LIGHT DIMMING RESISTOR
18 WARNING LIGHT DIMMING RESISTOR
19 WARNING LIGHT DIMMING RESISTOR
20 LEFT INBOARD DOOR OPEN SWITCH
21 RIGHT INBOARD DOOR OPEN SWITCH
22 LEFT INBOARD DOOR ACTUATING CYLINDER
23 RIGHT INBOARD DOOR ACTUATING CYLINDER
24 GEAR DOORS SWITCH
25 LEFT MAIN GEAR UPLOCK
26 RIGHT MAIN GEAR UPLOCK
27 LEFT MAIN GEAR UPLOCK SWITCH
28 RIGHT MAIN GEAR UPLOCK SWITCH
29 LEFT INBOARD DOOR FORWARD LOCK SWITCH
30 RIGHT INBOARD DOOR FORWARD LOCK SWITCH
31 LEFT INBOARD DOOR FORWARD LOCK
32 RIGHT INBOARD DOOR FORWARD LOCK
33 LEFT MAIN GEAR TORQUE ACTUATING CYLINDER
34 RIGHT MAIN GEAR TORQUE ACTUATING CYLINDER
35 LEFT MAIN GEAR DOWNLOCK SWITCH
36 RIGHT MAIN GEAR DOWNLOCK SWITCH
37 LEFT MAIN GEAR ACTUATING CYLINDER
38 RIGHT MAIN GEAR ACTUATING CYLINDER
39 LEFT MAIN GEAR STRUT
40 RIGHT MAIN GEAR STRUT
41 LEFT INBOARD DOOR AFT LOCK SWITCH
42 RIGHT INBOARD DOOR AFT LOCK SWITCH
43 LEFT INBOARD DOOR AFT LOCK
44 RIGHT INBOARD DOOR AFT LOCK
45 RIGHT MAIN GEAR TORQUE LINK SWITCH
46 WARNING SIGNAL GENERATOR
47 GEAR SELECTOR VALVE (LEFT SIDE)
48 GEAR DOORS SELECTOR VALVE (RIGHT SIDE)
49 LEFT THROTTLE POSITION SWITCH
50 RIGHT THROTTLE POSITION SWITCH
51 NOSE GEAR STRUT
52 NOSE GEAR ACTUATING CYLINDER
53 NOSE GEAR UPLOCK SWITCH
54 NOSE GEAR DOWNLOCK SWITCH
55 NOSE GEAR DOOR CLOSED SEQUENCE SWITCH
56 NOSE GEAR DOOR OPEN SEQUENCE SWITCH
57 AIRSPEED/ALTITUDE PRESSURE SWITCH
58 NOSE GEAR DOOR ACTUATOR
59 CAUTION & WARNING LIGHTS BRIGHT CIRCUIT BREAKER
60 LANDING GEAR WARNING CIRCUIT BREAKER
61 LANDING GEAR CONT, SAFETY & STEERING CIRCUIT BREAKER

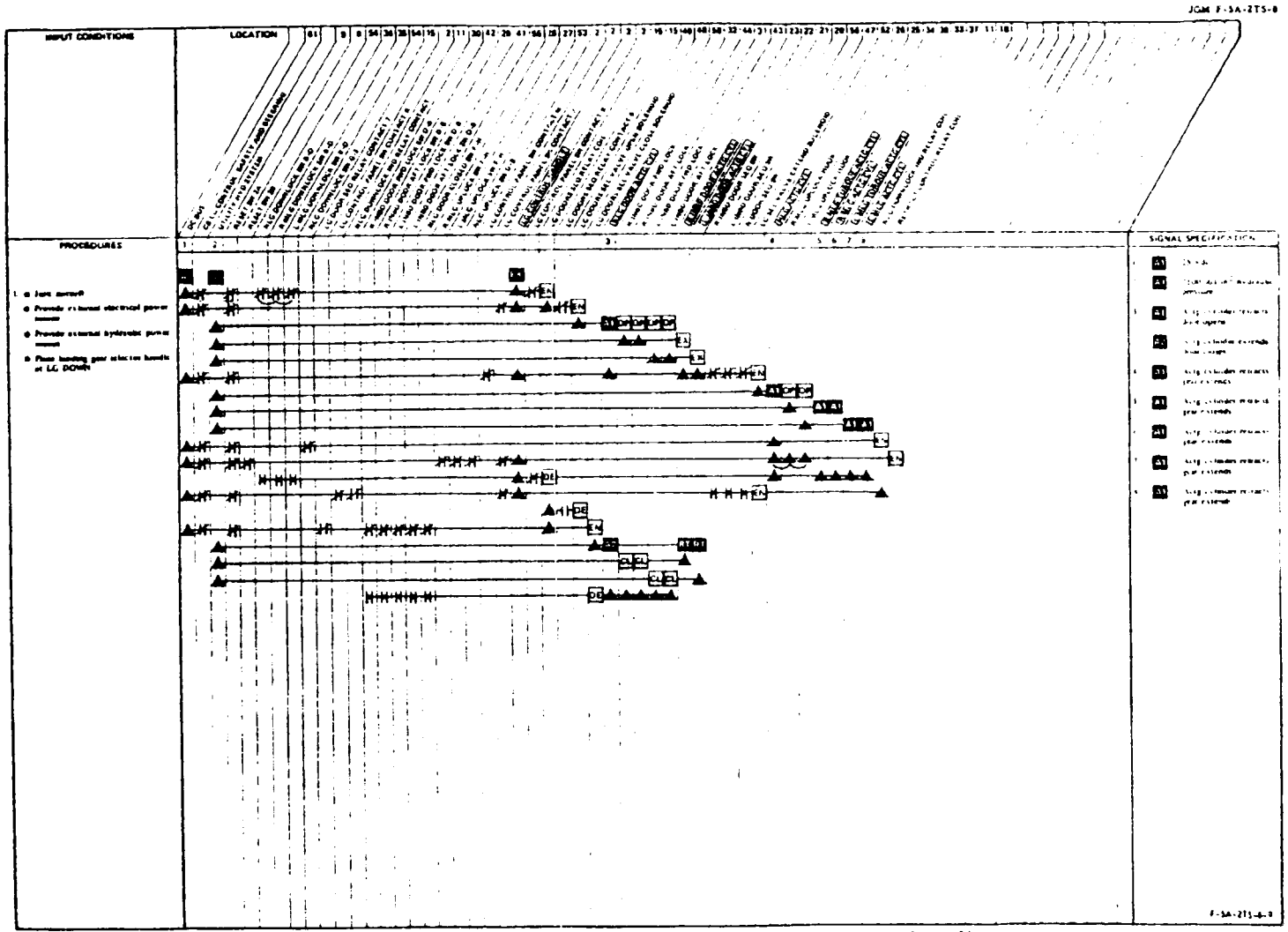
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MAINTENANCE DEPENDENCY CHART - LANDING GEAR RETRACTION (SHEET 1 OF 3)

Figure 6

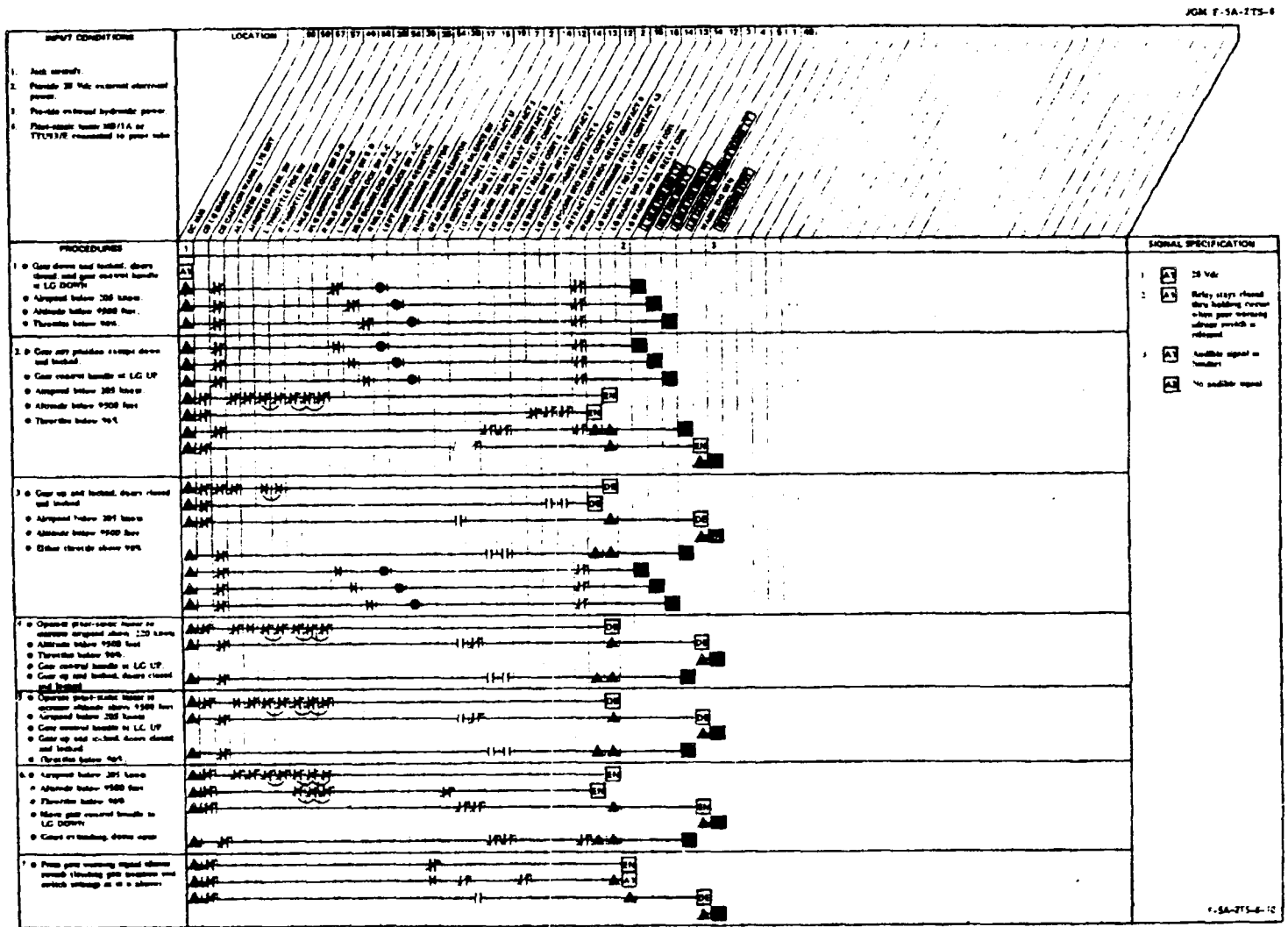
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MAINTENANCE DEPENDENCY CHART - LANDING GEAR EXTENSION (SHEET 2 OF 3)

Figure 9

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MAINTENANCE DEPENDENCY CHART – LANDING GEAR WARNING AND INDICATING SYSTEM (SHEET 3 OF 3)

Figure 6

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LANDING GEAR SYSTEM SYMPTOM-CAUSE TABLE

The following information is presented in table form to provide an easy-to-use reference for isolating landing gear problems, probable causes of the problem, and isolation and remedy procedures.

PROBABLE CAUSE	ISOLATION PROCEDURE	REMEDY
LANDING GEAR COMPLETELY INOPERATIVE IN NORMAL SYSTEM.		
Door selector valve defective.	Check selector valve solenoids for proper operation.	Replace selector valve.
Right main gear torque link switch out of adjustment or defective.	Make continuity check of switch to determine if defective.	Adjust switch, replace switch if defective.
		WARNING
Alternate release system switches not reset.	Check position of reset lever on Alternate release control mechanism.	Clear personnel and equipment from gear door areas. Reset Switches.
LANDING GEAR WILL NOT RETRACT.		
Landing gear selector valve defective.	Check selector valve solenoids for proper operation.	Replace selector valve.
Gear door-open sequence switches out of adjustment or defective.	Check all gear door-open sequence switches for proper operation.	Adjust switches, replace defective switches.
LANDING GEAR DOORS WILL NOT CLOSE.		
Gear uplock switch(es) out of adjustment or defective.	Check gear uplock switches for proper operation.	Adjust switches, replace defective switches.
Gear door selector valve defective.	Check selector valve solenoids for proper operation.	Replace selector valve.
		WARNING
Alternate release system switches not reset.	Check position of reset lever on Alternate release control mechanism.	Clear personnel and equipment from gear door areas. Reset switches.
LANDING GEAR WILL NOT EXTEND ON NORMAL SYSTEM.		
Landing gear selector valve extension solenoid defective.	Check selector valve extension solenoid for proper operation.	Replace selector valve.
ALTERNATE RELEASE SYSTEM WILL NOT EXTEND LANDING GEAR.		
Alternate release system not rigged properly.	Check Alternate release system rigging in accordance with section X.	Rig Alternate release system in accordance with section X.

PROBABLE CAUSE	ISOLATION PROCEDURE	REMEDY
LANDING GEAR SLOW IN RETRACTING OR EXTENDING.		
Landing gear adjustment or attach points not rigged properly or moving parts not properly lubricated.	Check landing gear for proper rigging and lubrication.	Adjust and lubricate landing gear as necessary
Restrictors in landing gear hydraulic lines obstructed.	Check restrictors in landing gear hydraulic lines for dirt or other obstructions.	Clean restrictor filters; replace damaged or defective restrictors.
ALTERNATE RELEASE SYSTEM SWITCHES WILL NOT RESET.		
Landing gear normal system not rigged properly.	Check gear control mechanism for proper rigging.	Rig landing gear system as necessary.
Alternate release system switches not properly adjusted.	Check Alternate release system switches in accordance with section X.	Rig switches in accordance with section X.

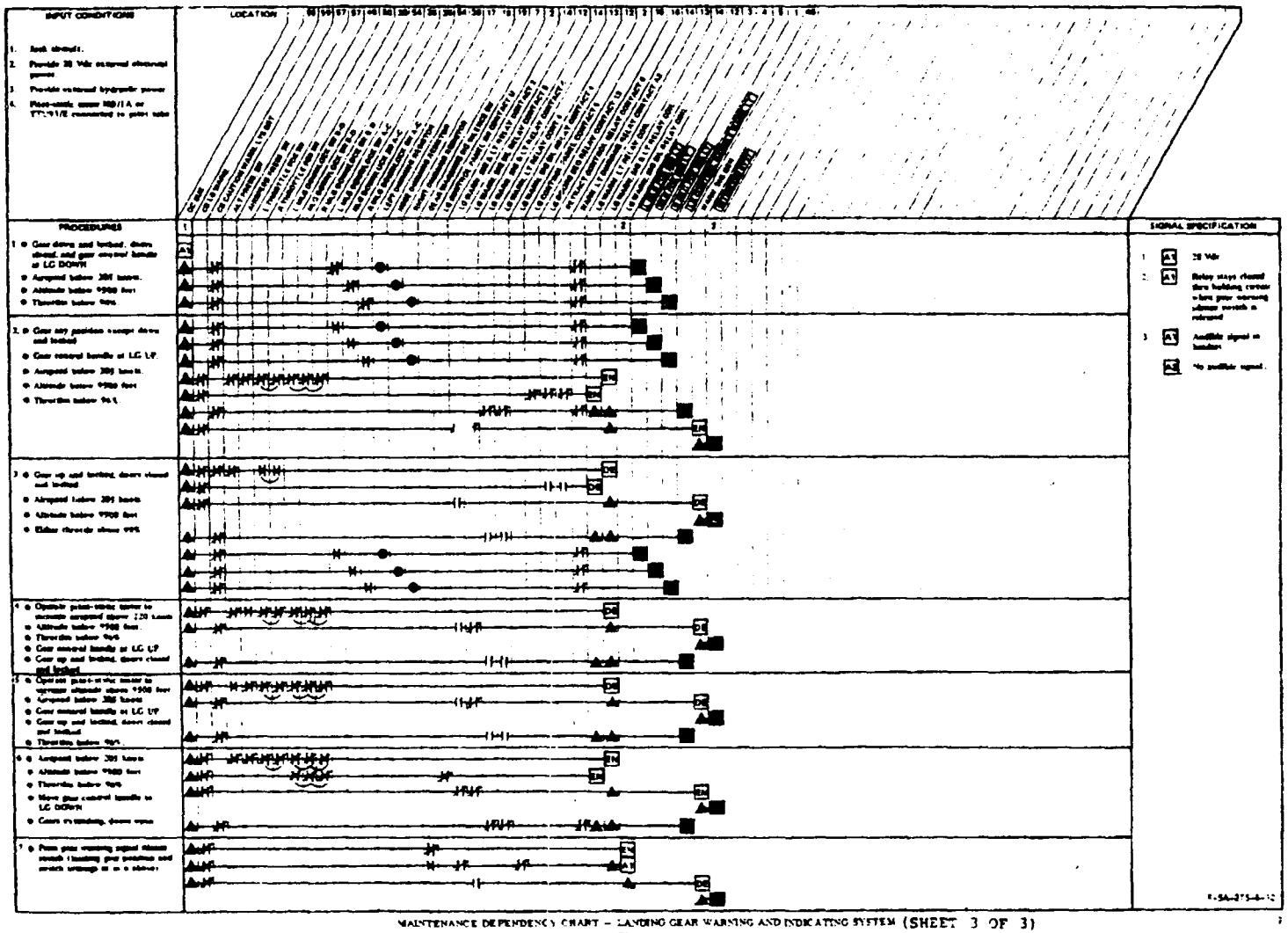
NOTE

Information pertaining to the landing gear alternate release system, nosewheel steering system, and wheel brake system was intentionally omitted from this sample

SUPPLEMENTAL DATA — LANDING GEAR SYSTEM

Figure 10

JGM F-5A-2TS-6



MAINTENANCE DEPENDENCY CHART – LANDING GEAR WARNING AND INDICATING SYSTEM (SHEET 3 OF 3)
Figure 9

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LANDING GEAR SYSTEM SYMPTOM-CAUSE TABLE

The following information is presented in table form to provide an easy-to-use reference for isolating landing gear problems, probable causes of the problem, and isolation and remedy procedures.

PROBABLE CAUSE	ISOLATION PROCEDURE	REMEDY
LANDING GEAR COMPLETELY INOPERATIVE IN NORMAL SYSTEM.		
Door selector valve defective.	Check selector valve solenoids for proper operation.	Replace selector valve.
Right main gear torque link switch out of adjustment or defective.	Make continuity check of switch to determine if defective.	Adjust switch; replace switch if defective.
		WARNING
Alternate release system switches not reset.	Check position of reset lever on alternate release control mechanism.	Clear personnel and equipment from gear door areas. Reset Switches.
LANDING GEAR WILL NOT RETRACT.		
Landing gear selector valve defective.	Check selector valve solenoids for proper operation.	Replace selector valve.
Gear door-open sequence switches out of adjustment or defective.	Check all gear door-open sequence switches for proper operation.	Adjust switches, replace defective switches.
LANDING GEAR DOORS WILL NOT CLOSE.		
Gear uplock switches out of adjustment or defective.	Check gear uplock switches for proper operation.	Adjust switches, replace defective switches.
Gear door selector valve defective.	Check selector valve solenoids for proper operation.	Replace selector valve.
		WARNING
Alternate release system switches not reset.	Check position of reset lever on alternate release control mechanism.	Clear personnel and equipment from gear door areas. Reset switches.
LANDING GEAR WILL NOT EXTEND ON NORMAL SYSTEM.		
Landing gear selector valve extension solenoid defective.	Check selector valve extension solenoid for proper operation.	Replace selector valve.
ALTERNATE RELEASE SYSTEM WILL NOT EXTEND LANDING GEAR.		
Alternate release system not rigged properly.	Check alternate release system rigging in accordance with section X.	Rig alternate release system in accordance with section X.

PROBABLE CAUSE	ISOLATION PROCEDURE	REMEDY
LANDING GEAR SLOW IN RETRACTING OR EXTENDING.		
Landing gear adjustment or attach points not rigged properly or moving parts not properly lubricated.	Check landing gear for proper rigging and lubrication.	Adjust and lubricate landing gear as necessary.
Restrictors in landing gear hydraulic lines obstructed.	Check restrictors in landing gear hydraulic lines for dirt or other obstruction.	Clean restrictor filters; replace damaged or defective restrictors.
ALTERNATE RELEASE SYSTEM SWITCHES WILL NOT RESET.		
Landing gear normal system not rigged properly.	Check gear control mechanism for proper rigging.	Rig landing gear system as necessary.
Alternate release system switches not properly adjusted.	Check alternate release system switches in accordance with section X.	Rig switches in accordance with section X.

NOTE

Information pertaining to the landing gear alternate release system, nosewheel steering system, and wheel brake system was intentionally omitted from this example

TRANSMISSION AND TAIL ROTOR POWER TRAIN

Applicable Serial Nos:

All UH-1N aircraft

Supplies:

One Maintenance In Progress tag
Three Circuit Breaker labels

Personnel Required: One

Rated pilot required to perform engine runup.

Special Tools and Test Equipment:

One Multimeter, AN/URM105

Equipment Conditions:

NOTE

Transmission must be serviced before beginning this activity.

1. Attach maintenance in progress tag to pilot's cyclic stick.
2. Open all AC circuit breakers (2). Open all DC circuit breakers (7).
3. Remove pylon access panels (5).
4. Check that there is no evidence of leakage.
5. Check that there is no evidence of leakage from grease seals.
6. Close and attach labels to following circuit breakers:

   CAUTION LIGHTS (1)
   TEMP IND ENG & XMSN (6)
   XMSN PRESSURE (3)

7. Place BAT switch (4) at ON.

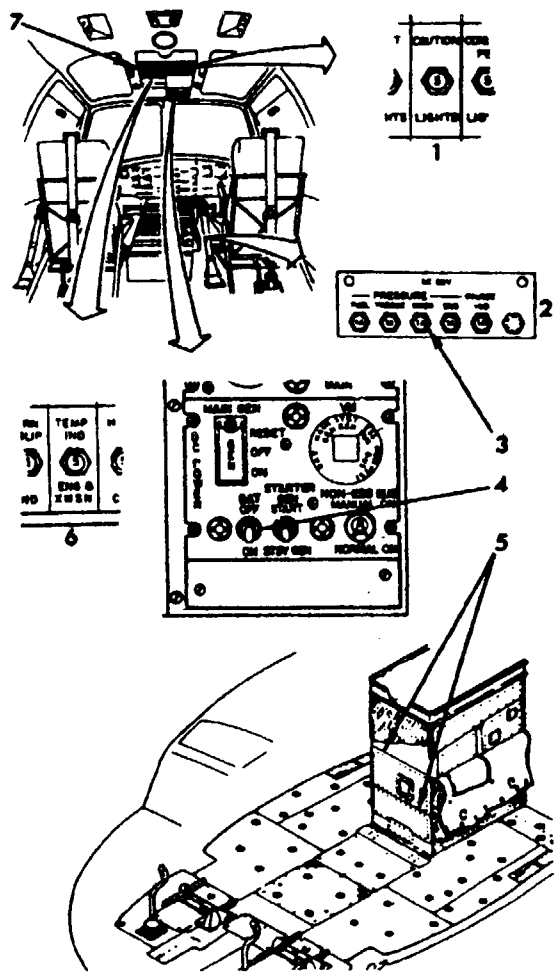
WARNING

Be sure area is clear of personnel and equipment before engine runup.

Engine runup must be performed by rated pilot.

CAUTION

Be sure all tools and loose miscellaneous equipment have been removed from transmission area.

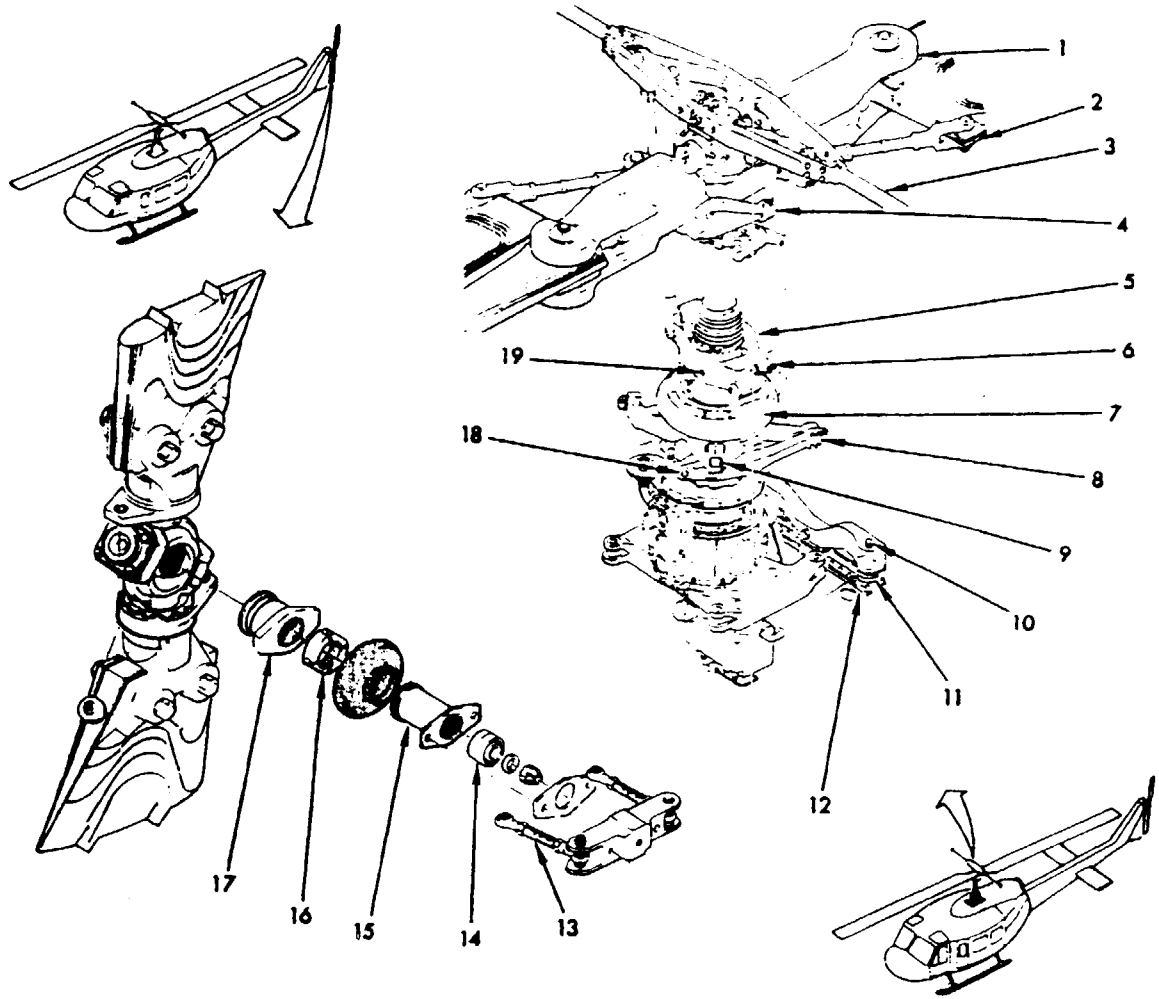


INPUT CONDITIONS PAGE - TRANSMISSION AND TAIL ROTOR POWER TRAIN

1

Figure 11

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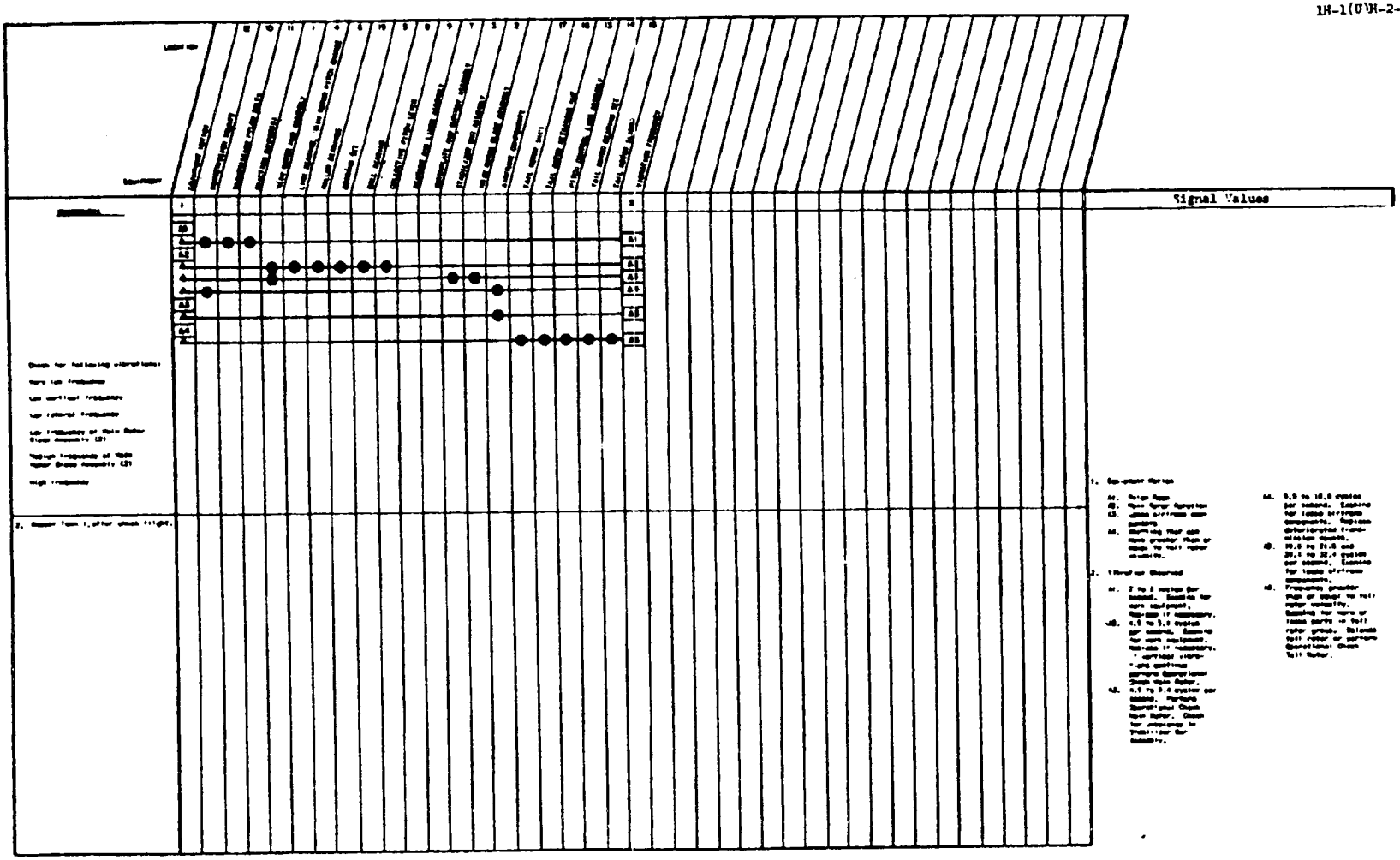


1. Main Rotor Hub Assembly
2. Main Rotor Blade Assembly
3. Stabilizer Bar Assembly
4. Main Rotor Pitch Change Link Bearing
5. Ball Bearing
6. Roller Bearing
7. Swashplate and Support Assembly
8. Collective Pitch Lever
9. Bearing and Liner Assembly
10. Transmission Pylon Bolts
11. Friction Dampers
12. Transmission Mounts
13. Pitch Control Link Assembly
14. Tail Rotor Bearing Set
15. Tail Rotor Slider
16. Tail Rotor Retaining Nut
17. Tail Rotor Shim
18. Bearing Liner
19. Bearing Set

LOCATOR PAGE - MAIN AND TAIL ROTOR

Figure 12

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MAINTENANCE DEPENDANCY CHART - MAIN AND TAIL ROTOR

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TRANSMISSION AND TAIL ROTOR POWER TRAIN

General

Transmission Oil Jets carry oil under pressure and deliver aimed sprays of oil on gears and bearings.

G Jet Assembly 1, sprays upper mast bearing, mast driving spline, and upper pinion bearings.

H Oil Auxiliary Jet 2, sprays spur gear and pinion bearings.

L Jet Assembly 3, sprays input bevel gears and delivers oil to Jet Assembly 6. (K)

S Jet Assembly 4, lubricates accessor, drive gears and tail rotor drive quill.

J Jet Assembly 5, lubricates input quill gears.

K Jet Assembly 6, receives oil from Jet 3. (L)

Sprays inboard bearing of input drive quill and lubricates bearings of freewheel coupling.

I Jet Assembly 7, lubricates bearings of internal gear quill which is driven by input drive gear quill.

F Jet Assembly 8, provides additional lubrication for upper mast bearing assembly and mast driving spline.

M Oil Manifold Relief Valve allows some oil to bypass oil manifold into transmission assembly.

O Thermobulb senses temperature for Transmission Oil Temperature Indicator (P).

N Thermoswitch senses temperature change for Master Caution panel.

Q Oil Manifold Inlet Hose delivers oil into oil manifold.

R Transmission Primary Oil Filter is an oil filter for transmission with a bypass valve for continued oil flow if transmission oil pump screens become clogged.

U Transmission Oil Pump Screen prevents foreign matter from getting into Oil Pump.

T Transmission Oil Level Sight Gage provides a visual inspection of oil level in transmission.

V Transmission Oil Cooler functions to cool transmission oil.

W Transmission External Oil Filter has a bypass valve set to open at 18 to 22 pounds per square inch gage to allow oil flow if transmission primary oil filter (R) becomes clogged.

D Transmission Oil Pressure Indicator receives pressure provided in pounds per square inch, from Transmission Oil Pressure Transmitter (E).

P Transmission Oil Temperature Indicator transmits oil temperature readings in degrees centigrade.

Engine Oil Temperature Indicator transmits temperature of engine oil at engine oil inlet in degrees centigrade.

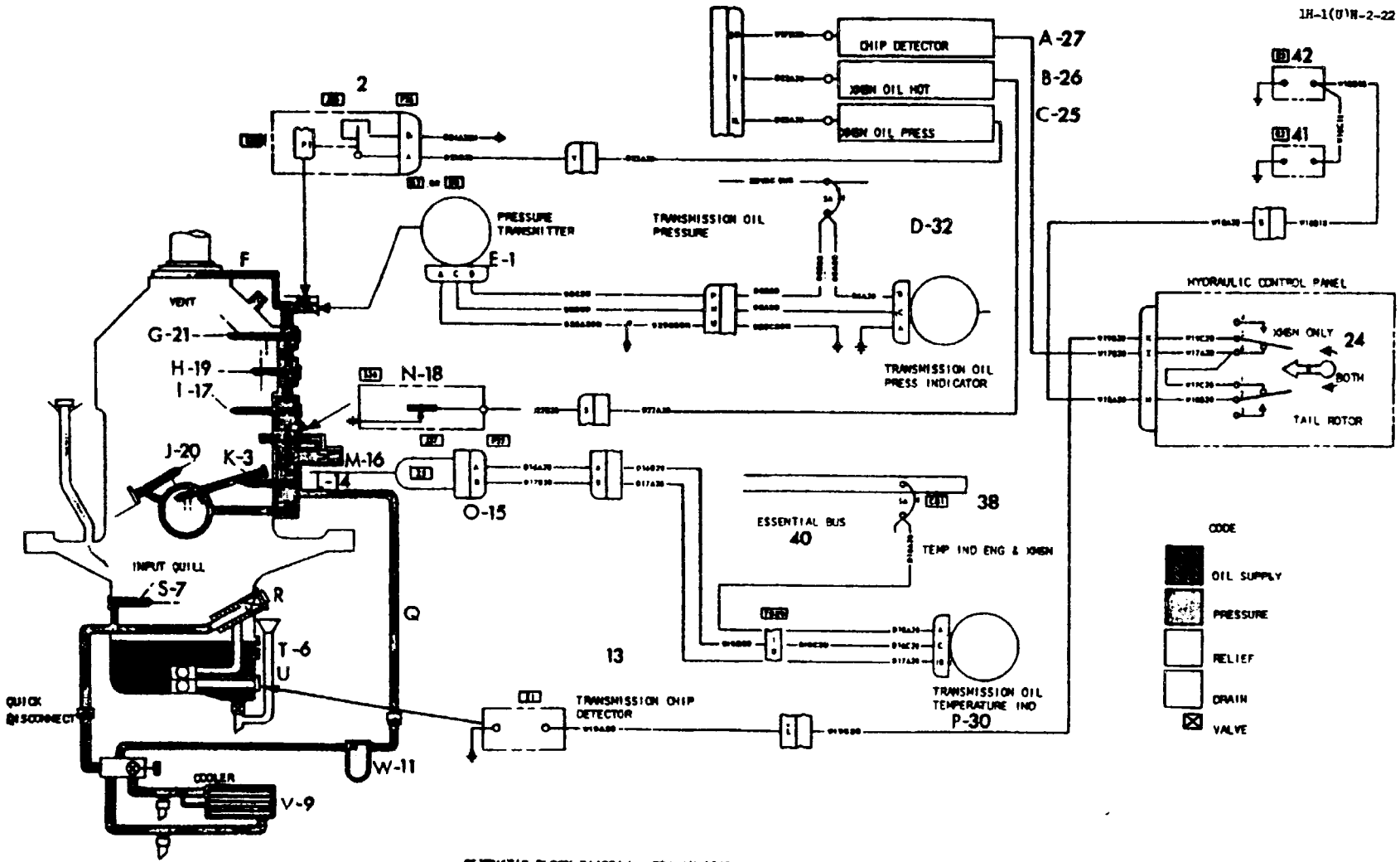
A CHIP DETECTOR Indicator Light provides means of inspecting for metal particles in transmission, intermediate and tail rotor gearboxes.

B XMSN OIL HOT Indicator Light indicates when transmission oil temperature reaches an unsafe limit.

C XMSN OIL PRESS Indicator Light indicates when transmission oil pressure drops below safe operating limits.

KEYED TEXT - TRANSMISSION AND TAIL ROTOR POWER TRAIN

Figure 14



SCHEMATIC BLOCK DIAGRAM - TRANSMISSION AND TAIL ROTOR POWER TRAIN

Figure 15

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Extreme Low Frequency vibration is limited to pylon rock. Pylon rocking two to three cycles per second is inherent with rotor, mast, and transmission system.

To keep vibration from reaching noticeable levels, transmission mount dampening is incorporated to absorb rocking.

A quick check of dampening system may be performed by pilot while in a hover maneuver.

An abnormal continuation of rock during a check or during normal flight, is an indication that something is wrong with transmission mounts or dampers.

Low Frequency vibrations, one or two vibrations per revolution, are caused by Main Rotor. One vibration per revolution occurs when one blade develops more lift at a given point than the other blade at same point.

Low Frequency Vertical vibrations are felt in a low power descent at moderate airspeeds (60 to 70 knots) because of a difference in blade lift.

Verticals felt in forward flight, that get worse as airspeed increases, are usually due to one blade developing more lift with increased speed, than the other blade.

Associated with one vibration per revolution vertical is the intermittent one vibration per revolution vertical. This is a vibration initiated by a gust effect causing a momentary increase of lift in one blade giving one vibration per revolution.

Momentary vibration is normal but it is undesirable if picked up by rotating collective controls and fed back to rotor to cause several cycles of one vibration per revolution.

Sometimes during steep turns, one blade will go out of track and cause a hard one vibration per revolution vertical. This is usually caused by too much differential tab in blades.

Low Lateral Frequency vibration occurs when a rotor blade, or rotor component is out of balance and causes one vibration per revolution lateral vibrations.

This vibration is usually felt as a vertical, due to rolling motion. It causes pilot's seat to bounce up, while copilot's seat goes down.

An unusually severe lateral can be felt as a definite sideward motion as well as a vertical motion.

Lateral vibrations occurring to an unbalance in rotor are of two types, spanwise and cordwise.

Spanwise unbalance is caused by one hub and blade assembly being heavier than the other.

Cordwise unbalance is caused by more weight toward trailing edge of one blade than the other.

Third cause of lateral vibrations is improper balancing of stabilizer bar assembly.

Lateral vibrations are usually felt in a hover and in descending moderate airspeed turns, and tend to disappear in forward flight.

An out-of-ground hover is usually the best place to feel a lateral. A reduction to 6000 RPM will often make lateral vibration more prominent.

To correct one vibration per revolution lateral vibration, begin by determining if one blade is heavier than another.

Should vibration still be present after Operational Check of Main Rotor, Hub and Blade Assembly should be removed.

Check for alignment of Hub and Blade Assembly as well as balancing of stabilizer bars.

Low Frequency two vibrations per revolution vibrations are inherent with two bladed rotor systems. A low level vibration is always present.

A marked increase over normal two vibrations per revolution level can be caused by two basic factors. A loss of dampening or absorption capability or an actual increase in two vibrations per revolution level of rotor.

Loss of dampening can be caused by such factors as deteriorated transmission mounts or lift link bushings, or an airframe component loosening up and vibrating in sympathy with inherent two vibrations per revolution vibration.

An increase in two vibrations per revolution level of rotor itself can be caused by worn or loose parts in rotor hub or looseness in rotating controls.

Medium Frequency Vibrations at frequencies of four vibrations per revolution and six vibrations per revolution are also inherent vibrations.

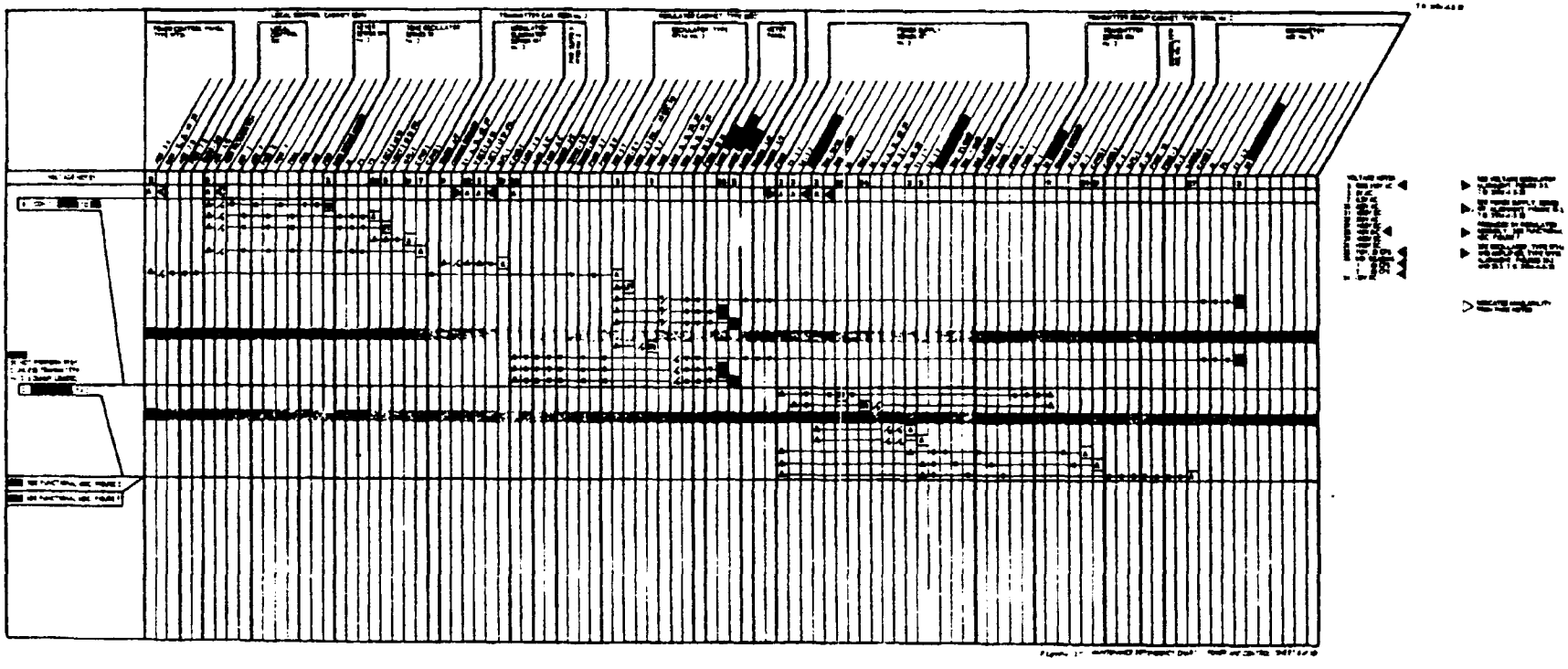
An increase of these vibrations is caused by a change in capability of fuselage to absorb vibration, or a loose airframe component.

Changes in fuselage vibration absorption can be caused by such things as fuel level, external stores, structural damage, structural repairs, internal loading, or gross weight.

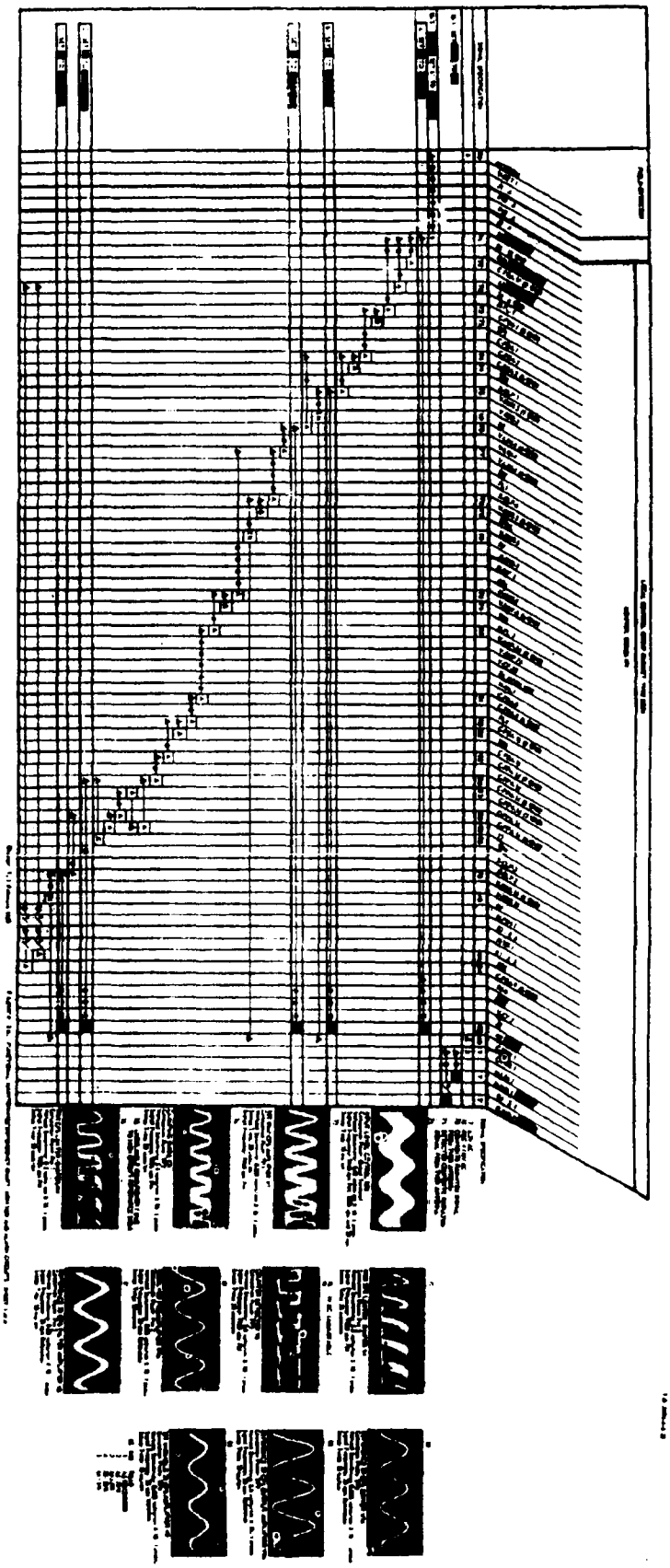
Vibration is felt as a rattling in aircraft structure.

SUPPLEMENTAL Data - MAIN AND TAIL ROTOR (2 of 3)

Figure 16



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